

MORARJI DESAI RESIDENTIAL SCHOOL

SC-32 Bahaddurghatta (Kogunde) · Chitradurga District — 577519

Lesson Plans

Level 2 · Class 7 · 72 lessons · 40-minute periods

PREPARED BY

Hoysala T, Computer Science Teacher

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LESSON 1.01 · CONCEPT · 40 MINUTES

Network Types

I CAN Define computer network and explain its importance · Identify different types of networks with examples · Compare personal area network, local area network, and wide area network ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Students, imagine you are in the dormitory and you want to share a photo from your phone to your friend's laptop. How would you do it? Think about all the ways information travels between devices.	Students mention Bluetooth, Wi-Fi, cables, and USB drives. They discuss which method is fastest and why.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Look at the network diagram on the board. You can see three computers connected in the classroom. Now look at the bigger diagram showing connections between our school and other schools in the district. What differences do you notice in the size and distance of these connections?	Students observe that classroom network is small and short distance while district network connects many buildings over long distance. They note more devices in larger network.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A computer network is a group of devices connected together to share data and resources. Networks are classified by size. PAN covers one person like phone to headphones. LAN covers one building like our school. MAN covers a city. WAN covers countries or the whole world like the Internet.	Students take notes on network types with examples. They create a comparison table showing range, speed, and examples for PAN, LAN, MAN, and WAN.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Think about residential school system. We have multiple schools across Karnataka. What type of network would connect all residential schools together? What type of network exists inside each school computer lab?	Students identify that inter-school connection is WAN while school lab network is LAN. They draw a simple network diagram showing residential school connections.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Match each network type to its correct example. Write one example from your daily life for each network type.	Students correctly match PAN to phone headset, LAN to computer lab, MAN to city network, and WAN to Internet. They provide personal examples like smartwatch to phone.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Network diagram poster · Chart paper · Markers · Textbook Chapter 1 · Computer with projector ·

DIFFERENTIATION **App** Provide pre-made flashcards with network definitions and matching pictures for visual... **On** Students complete standard matching and labeling activities from textbook **Adv** Challenge students to research and present about satellite networks and their use in... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Tonight ask your hostel warden what type of network the school uses. Spiral: also recall one fact from last week.

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LESSON 1.02 · CONCEPT · 40 MINUTES

LAN and WAN

I CAN Explain the difference between LAN and WAN in detail · Describe advantages and disadvantages of each network type · Identify real-world examples of LAN and WAN in Karnataka ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE 5 min	Close your eyes and imagine two scenarios. First, you are sharing a file with the student sitting next to you in class. Second, you are sending a message to your friend in another residential school fifty kilometers away. What is different about these two situations?	Students discuss proximity, speed, and complexity. They realize short distance sharing is simpler than long distance communication.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE 10 min	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Examine this comparison table on the board. LAN column has short distance, high speed, private ownership. WAN column has long distance, lower speed, public or rented. Now look at the network map showing our school LAN connecting to the Internet which is the biggest WAN.	Students identify that LAN connects devices in one location while WAN connects distant locations. They note LAN is faster and more secure than WAN.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN 10 min	LAN or Local Area Network connects devices within a small area like a building or campus. It uses Ethernet cables or Wi-Fi. Speed is high from one hundred Mbps to ten Gbps. WAN or Wide Area Network connects devices over large distances using telephone lines, fiber optics, or satellites. Internet is the largest WAN in the world.	Students create a Venn diagram comparing LAN and WAN. They list five features of each and identify two features common to both.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE 10 min	Our residential school has a LAN in the computer lab. When we connect to the Internet we are joining a WAN. Design a simple network for a residential school campus showing the computer lab, library, staff room, and hostel all connected.	Students sketch a campus LAN with labeled devices. They show the Internet connection as a cloud outside the campus boundary.	<i>Peer tick against Success Criteria</i>
EVALUATE 5 min	Answer these questions. What is the full form of LAN? Give two advantages of LAN over WAN. Name one WAN you use every day. Draw a simple LAN diagram.	Students answer correctly showing understanding of definitions, advantages, and real-world examples. Diagrams show proper device connections.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Ethernet cable sample · LAN diagram handout · Textbook page 12 · Whiteboard and markers · Computer lab access ·

DIFFERENTIATION **App** Provide sentence starters for comparing LAN and WAN with guided word banks **On** Students complete standard comparison chart and answer textbook questions **Adv** Advanced students research metropolitan area networks and describe how they differ from... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Count how many devices in your dormitory room connect to the school Wi-Fi. Spiral: also recall one fact from last week.

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LESSON 1.03 · CONCEPT · 40 MINUTES

Internet Services

I CAN List and explain major Internet services available to users · Differentiate between web browsing, email, and file transfer services · Describe how Internet services help in education at residential schools ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	What do you do when you first open a browser on the school computer? Think about all the different things you can do on the Internet in one hour. List as many activities as you can think of.	Students brainstorm activities like watching videos, sending messages, searching for information, downloading files, reading news, and playing games online.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Look at these screenshots on the board. The first shows a website browser. The second shows an email inbox. The third shows a file download. The fourth shows a video call. For each screenshot, identify what Internet service is being used.	Students identify web browsing, email, file transfer, and video conferencing. They note that each service serves a different purpose.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	The Internet provides many services. World Wide Web lets you access websites and information. Email lets you send and receive electronic messages. FTP lets you transfer files between computers. Video conferencing lets you have face-to-face conversations online. Social media lets you connect with friends and share content.	Students create a mind map with Internet services at the center branching out to each service with examples. They write one advantage of each service.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Think about a day in residential school without Internet services. How would your learning be affected? Which Internet service is most important for your studies and why? Present your reasoning to the class.	Students argue for different services based on educational value. Some choose web search for research, others choose email for communication with teachers, and some choose video for learning.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Draw a table with columns Service, Purpose, and Example. Fill in at least five Internet services with their purposes and real examples you have used.	Students create accurate tables showing web browsing for information, email for communication, FTP for file sharing, streaming for entertainment, and cloud storage for backup.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Screenshot prints of different services · Projector · Textbook · Worksheet with service table · Internet access for demo ·

DIFFERENTIATION **App** Provide a partially filled table with picture prompts for each Internet service **On** Students complete the standard activity identifying and describing five Internet services **Adv** Challenge students to evaluate and rank Internet services by importance for rural... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List all Internet services you used today before sleeping.
Spiral: also recall one fact from last week.

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LESSON 1.04 · CONCEPT · 40 MINUTES

Email Advanced

I CAN Compose professional emails with proper structure and etiquette · Use email features like CC, BCC, and attachments effectively · Understand email security and avoid phishing threats ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine your school principal wants to send an important announcement to all our school teachers and also wants to copy the district education officer. What email features would help in this situation? Think about who should see the email and who should remain hidden.	students discuss CC for visible recipients, BCC for hidden recipients, and the importance of professional email communication in official settings.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Examine these three emails on the board. The first is informal with no subject and spelling errors. The second is formal with proper greeting and structure. The third has a suspicious link and unknown sender. Which email would you trust and why?	students identify that formal structured emails are more professional and trustworthy. They notice phishing signs like suspicious links and unknown senders in the third email.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Professional emails have a clear subject line, proper greeting like Dear or respected, concise body paragraphs, and a formal closing with your name and class. CC or carbon copy sends the email to additional visible recipients. BCC or blind carbon copy sends the email without other recipients knowing. Attachments let you include files like documents or images.	students practice writing a formal email to a librarian requesting book reservation. They use CC to include their class teacher and BCC to inform the warden secretly.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Write an email to the district education officer requesting new computers for the residential school lab. Include your class teacher in CC and the principal in BCC. Attach a list of required items as a separate document.	students compose complete professional emails with all elements. They verify correct use of CC and BCC and practice attaching files using the school computers.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Review the email you wrote. Check for proper subject line, greeting, body structure, closing, and correct use of CC and BCC. Rate your email on a scale of one to five for professionalism.	students self-evaluate their emails against a rubric. They identify areas for improvement and revise their drafts for clarity and professionalism.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Email templates · Phishing example printouts · Computer with email client · Rubric for email assessment · Textbook ·

DIFFERENTIATION **App** Provide email templates with fill-in-the-blank sections for subject, greeting, and closing **On** Students write complete emails following the standard professional email format **Adv** Advanced students analyze real phishing emails and create a detection guide for classmates **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a draft email to your parents describing your week at residential school. Spiral: also recall one fact from last week.

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LESSON 1.05 · CONCEPT · 40 MINUTES

Network Topologies

I CAN Define network topology and explain its importance · Describe star, bus, ring, and mesh topologies with diagrams · Identify advantages and disadvantages of each topology ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at how the desks are arranged in our classroom. Some are in rows, some are in a circle, and some are grouped together. If each desk represented a computer, how would you connect them? Think about the shape of connections.	students suggest different connection patterns like straight lines, circles, and web-like patterns. They realize connection shape affects communication flow.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Use these yarn pieces and cards to create four different network shapes on the floor. First arrange them in a straight line like a bus. Then arrange them in a circle like a ring. Next arrange them around one central card like a star. Finally connect every card to every other card like a mesh.	students physically create topology models. They observe that mesh uses the most yarn and star uses a central hub. Bus is simplest to create.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Topology is the arrangement of devices in a network. Bus topology connects all devices to one central cable like beads on a string. Star topology connects all devices to one central hub like spokes of a wheel. Ring topology connects devices in a closed loop like a necklace. Mesh topology connects every device to every other device like a spider web.	students draw diagrams of all four topologies and label each component. They create a comparison table listing cost, reliability, and troubleshooting difficulty for each.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Our school computer lab uses one of these topologies. Observe the cable connections in the lab and identify the topology used. Explain why this topology was chosen for our lab based on its advantages.	students identify star topology as most common in labs because if one cable fails others continue working. They discuss how central switch makes troubleshooting easy.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Draw all four network topologies from memory. Label the components of each. Write two advantages and two disadvantages of star topology compared to bus topology.	students produce accurate diagrams with correct labels. They articulate clear comparisons between topologies showing understanding of trade-offs.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Yarn pieces · Index cards · Scissors · Lab topology diagram · Textbook · Floor space for models ·

DIFFERENTIATION **App** Provide pre-cut cards and guided instructions for building each topology model step by... **On** Students create topology models and diagrams following standard textbook descriptions
Adv Advanced students research hybrid topologies and explain how combining topologies... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the lights in your dormitory corridor. Spiral: also recall one fact from last week.

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LESSON 1.06 · CONCEPT · 40 MINUTES

TCP/IP Basics

I CAN Explain the purpose of TCP and IP in network communication · Describe how data is broken into packets and reassembled · Understand the four-layer TCP/IP model ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine you need to send a large textbook to your friend in another residential school by courier. The courier company has small boxes. How would you pack the book? What information would you write on each box to ensure it reaches the correct destination?	students discuss breaking the book into sections, labeling each box with sender and receiver address, and trusting the courier to deliver all boxes correctly.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Use these Lego blocks to simulate data transmission. The tower represents a large file. Break it into smaller stacks like packets. Label each stack with a number. Now arrange them at the destination in the correct order using the numbers.	students experience how large data is divided into packets, transmitted separately, and reassembled at destination. They notice packets may arrive out of order.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	TCP or Transmission Control Protocol breaks data into small packets, numbers them, and ensures all packets arrive correctly at the destination. If any packet is lost TCP requests retransmission. IP or Internet Protocol handles addressing and routing. It puts the sender and receiver addresses on each packet so routers know where to send them.	students create a flow diagram showing data being broken into packets at source, traveling through routers, and being reassembled at destination. They label TCP and IP functions.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	The TCP/IP model has four layers. Application layer handles email, web browsing, and file transfer. Transport layer handles TCP and UDP. Internet layer handles IP addressing. Network access layer handles physical transmission. Match each layer to its function.	students create a four-layer diagram and place example activities in each layer. They discuss how layers work together like different departments in a company.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Explain in your own words how TCP ensures reliable data transmission. Describe the role of IP in delivering packets to the correct destination. Draw the four-layer TCP/IP model.	students demonstrate understanding through clear explanations and accurate diagrams. They connect concepts to real Internet usage.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Lego blocks · Numbered labels · Paper for packet simulation · Textbook diagrams · Whiteboard · Colored markers ·

DIFFERENTIATION **App** Provide a simplified packet diagram with pre-labeled layers and guided matching activities **On** Students complete standard labeling and description activities from the textbook **Adv** Advanced students research UDP protocol and compare its speed versus TCP reliability... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about a message you sent on your phone today.
Spiral: also recall one fact from last week.

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LESSON 1.07 · CONCEPT · 40 MINUTES

Network Devices

I CAN Identify common network devices and their functions · Explain the role of routers, switches, and modems in a network · Describe how network devices work together to enable communication ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about a post office. Letters arrive from many places and need to be sorted and sent to the correct addresses. A network works similarly with special devices. Can you guess what devices help sort and direct data in a network?	students guess devices like routers, switches, and modems. They draw parallels between post office sorting and network device functions.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Examine the network equipment in our computer lab. Identify the modem that connects to the Internet line, the router that distributes the connection, and the switches that connect individual computers. Trace the cable from the wall socket to a computer.	students physically locate and trace network devices. They observe the connection path from Internet line through modem to router to switch to computer.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A modem converts digital signals from your computer to analog signals for telephone lines and back. A router directs data packets between different networks like from your school network to the Internet. A switch connects multiple devices within a LAN and directs data to the correct device. An access point provides wireless connectivity. A hub broadcasts data to all devices unlike a switch which sends only to the intended device.	students create a device diagram showing how each device connects in sequence. They write the function of each device and draw signal flow arrows.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Design a complete network for a new residential school building with twenty computers, a printer, and Internet access. Choose and place the correct network devices. Explain why each device is needed and where it should be located.	students propose modem for Internet conversion, router for distribution, switches for connecting computers, and print server for shared printing. They justify placement based on cable length and signal strength.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Match each device to its function. Router directs packets between networks. Switch connects devices in LAN. Modem converts signals. Access point provides Wi-Fi. Draw the device connection sequence.	students correctly match all devices and draw accurate connection diagrams showing the path from Internet to end device.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Modem sample · Router image · Switch diagram · Lab network layout · Cable samples · Textbook ·

DIFFERENTIATION	App Provide device cards with pictures and simplified function descriptions for matching... Advanced students research load balancers and firewalls and their roles in enterprise...	On Students complete standard device identification and function description tasks	Adv Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Find the Wi-Fi router in your hostel block. Spiral: also recall one fact from last week.		

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LESSON 1.08 · CONCEPT · 40 MINUTES

Wireless Networks

I CAN Explain how wireless networks use radio waves to transmit data · Describe Wi-Fi standards and their differences · Set up and secure a basic wireless network connection ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	How does music travel from your phone to your Bluetooth earphones without any wires? How does the Internet reach your laptop in the hostel without a cable plugged into it? What invisible force carries these signals?	students discuss radio waves, Bluetooth, and Wi-Fi. They wonder how information travels through air without physical connection.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Use the school Wi-Fi settings to view available networks. Note the network name, signal strength, and security type. Now compare with a neighbor's phone showing different available networks. What networks appear on both devices?	students observe that both devices see the same school Wi-Fi network. They notice signal strength varies based on distance from the access point.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Wireless networks use radio waves to transmit data between devices. Wi-Fi follows IEEE 802.11 standards. 802.11n offers speeds up to six hundred Mbps. 802.11ac offers speeds up to three point five Gbps. 802.11ax or Wi-Fi 6 offers even faster speeds with better performance in crowded areas. Wi-Fi range typically covers thirty to one hundred meters indoors.	students compare Wi-Fi standards in a table noting speed, frequency, and range. They understand why newer standards are better for streaming and gaming.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Test the Wi-Fi signal strength at different locations in the school campus. Measure signal at the computer lab, library, dining hall, and hostel. Create a signal strength map showing the best and weakest coverage areas.	students walk around campus with devices recording signal bars. They identify dead zones and suggest where additional access points could improve coverage.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	List three advantages of wireless networks over wired networks. Describe two security measures used to protect Wi-Fi networks. Explain why Wi-Fi signal becomes weaker through walls.	students explain convenience and mobility advantages. They describe WPA encryption and password protection. They understand that physical barriers block radio waves.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES School Wi-Fi access · Smartphones for signal testing · Signal strength apps · Coverage map template · Textbook ·

DIFFERENTIATION **App** Provide a pre-made signal strength chart with picture icons for recording measurements **On** Students conduct standard Wi-Fi coverage testing and record results in a table **Adv** Advanced students research mesh Wi-Fi systems and present how they eliminate dead zones... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Check the Wi-Fi signal strength in your dormitory room tonight. Spiral: also recall one fact from last week.

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LESSON 1.09 · CONCEPT · 40 MINUTES

Internet Protocols

I CAN Define Internet protocols and explain their purpose · Describe HTTP, HTTPS, FTP, and SMTP protocols with examples · Understand the difference between secure and non-secure protocols ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	When you visit a website you see either http or https in the address bar. What do these letters mean? Why do some websites show a green lock symbol while others show a warning? Think about the rules that govern Internet communication.	students notice the difference between http and https websites. They observe lock symbols on banking sites and warning signs on insecure sites.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Open five different websites on the school computer. Check the address bar of each. Categorize them into http and https groups. Now try to access an FTP site to download a free software. Observe how the connection process differs from regular web browsing.	students find most educational sites use https while some simple sites still use http. They experience FTP as a different type of file access interface.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Protocols are rules that govern communication. HTTP or Hypertext Transfer Protocol sends data in plain text without encryption. HTTPS adds SSL or TLS encryption making it secure. FTP or File Transfer Protocol is used for uploading and downloading files between servers and computers. SMTP or Simple Mail Transfer Protocol is used for sending emails. POP3 and IMAP are used for receiving emails.	students create a protocol chart with columns for name, full form, purpose, security level, and port number. They highlight that HTTPS is essential for secure transactions.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Simulate a web request. One student acts as the browser sending an HTTP request. Another acts as the server responding with a web page. A third student acts as a hacker trying to read the message. Use plain text first then use encoded words to simulate HTTPS encryption.	students experience the difference between unencrypted and encrypted communication. The hacker can easily read plain text but cannot decode encrypted messages.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Match each protocol to its function. HTTP for web browsing, HTTPS for secure browsing, FTP for file transfer, SMTP for sending email. Explain why online shopping requires HTTPS instead of HTTP.	students correctly match protocols and explain that HTTPS protects credit card and personal information during online transactions.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Computer with browser · FTP client demo · Protocol comparison chart · SSL certificate example · Textbook ·

DIFFERENTIATION **App** Provide a protocol matching worksheet with pictures and simplified descriptions **On** Students complete standard protocol identification and comparison activities **Adv** Advanced students research DNS protocol and explain how website names translate to IP... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Check the address bar of any website you visit tonight.
Spiral: also recall one fact from last week.

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LESSON 1.10 · CONCEPT · 40 MINUTES

Network Security

I CAN Identify common network security threats and attacks · Describe protection methods including passwords, firewalls, and antivirus · Practice safe browsing habits and recognize phishing attempts ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Your friend tells you that someone hacked into their email and read all their private messages. How do you think this happened? What would you do to prevent it from happening to you?	students discuss weak passwords, suspicious links, and public Wi-Fi dangers. They realize security requires both technology and personal vigilance.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Examine these examples of phishing emails and fake websites on the board. Some look very real with school logos and official language. Others have spelling errors and suspicious links. Identify the red flags in each example that reveal it is fake.	students spot suspicious email addresses, urgent language demanding immediate action, misspelled URLs, and requests for passwords. They learn to verify sender identity.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Network security threats include viruses that damage files, worms that spread across networks, phishing that steals passwords through fake sites, and denial of service attacks that crash websites. Protection includes strong passwords with letters numbers and symbols, firewalls that block unauthorized access, antivirus software that detects malware, and HTTPS encryption that protects data in transit.	students create a security guide listing five threats and five protection methods. They practice creating strong passwords and identifying phishing signs.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Conduct a security audit of the school computer lab. Check if antivirus is updated, firewall is enabled, passwords are strong, and browsers are current. Write a report with recommendations for improving security.	students inspect lab computers and find that some have weak passwords like one two three four five six. They recommend password policies and regular updates.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Classify each scenario as safe or risky. Using HTTPS for banking. Clicking links in unknown emails. Updating antivirus regularly. Sharing passwords with friends. Downloading software from official sites only.	students correctly identify risky behaviors. They explain why sharing passwords and clicking unknown links are dangerous practices.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Phishing email samples · Password strength checker · Antivirus software demo · Security checklist · Textbook ·

DIFFERENTIATION **App** Provide a picture-based security matching activity with threat icons and solution icons **On** Students complete standard threat identification and protection method matching **Adv** Advanced students research encryption algorithms and present how AES protects data **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Check if your phone has a screen lock. Spiral: also recall one fact from last week.

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LESSON 1.11 · CONCEPT · 40 MINUTES

Cloud Services

I CAN Define cloud computing and explain its benefits · Identify popular cloud services like Google Drive and OneDrive · Compare local storage versus cloud storage advantages and limitations ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You have five hundred photos on your phone but the storage is full. You could delete some photos or move them somewhere that never gets full and you can access from any device. Where would you move them?	students think of external hard drives and cloud storage. They realize cloud storage can be accessed from anywhere with Internet.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Log into the school Google account and explore Google Drive. Notice how much free storage is available. Create a folder, upload a text file, and share it with a classmate. Observe how the file appears on your classmate's device immediately.	students experience instant file synchronization across devices. They see that cloud storage requires Internet but provides accessibility and sharing features.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Cloud computing means storing and accessing data and programs over the Internet instead of your computer's hard drive. Cloud storage services include Google Drive with fifteen GB free, Microsoft OneDrive with five GB free, and Dropbox with two GB free. Advantages include access from anywhere, automatic backup, easy sharing, and no hardware to carry. Disadvantages include need for Internet, privacy concerns, and storage limits.	students create a comparison table of cloud storage services with columns for free space, features, and best use cases. They understand trade-offs between convenience and control.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Imagine residential school wants to move all student records to the cloud. What are the benefits and risks? Design a plan for safe cloud migration including what data to move, what security measures to implement, and how to train staff.	students propose benefits like teacher access from anywhere and automatic backup. They identify risks like data breaches and propose encryption and access controls.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Answer these questions. What is cloud storage? Name two cloud services. List two advantages over USB drives. Explain when local storage is better than cloud storage.	students define cloud storage correctly, name services, list accessibility advantages, and explain that large files or offline work need local storage.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES

DIFFERENTIATION **App** Provide step-by-step picture guides for uploading files to Google Drive with screenshots **On** Students complete standard cloud storage comparison and file upload activities **Adv** Advanced students research cloud computing models like SaaS, PaaS, and IaaS with real... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Check how much storage is left on your phone. Spiral: also recall one fact from last week.

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LESSON 1.12 · CONCEPT · 40 MINUTES

Networking Project

I CAN Apply networking knowledge to design a complete school network · Present network solutions with proper documentation · Evaluate network designs based on cost, speed, and reliability ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Our school wants to build a new computer lab with thirty computers, a printer, Internet access, and Wi-Fi for the whole campus. You have been hired as the network consultant. What will you need to plan and design?	students brainstorm requirements including devices, cables, topologies, security, and budget. They realize networking projects need careful planning.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Concrete (CPA): Use yarn and node cards on the floor to physically build network topologies (bus, ring, star, mesh) before drawing abstract diagrams. Examine the network designs created by previous batches. Compare three different designs for a ten-computer lab. One uses only wireless, one uses only wired, and one uses a hybrid approach. Evaluate which design is best and why.	students identify that hybrid design offers best balance of speed and flexibility. They note wireless is convenient but wired is more reliable for heavy use.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A network project requires planning several components. Hardware includes computers, servers, switches, routers, and cables. Software includes operating systems and network management tools. Security includes firewalls, antivirus, and access controls. Documentation includes network diagrams, IP address allocation, and maintenance schedules. Budget includes one-time setup cost and annual maintenance cost.	students create a project checklist with all components. They understand that each component must be planned before implementation begins.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Work in teams to design a complete network for a new residential school wing with a library, computer lab, and staff room. Create a network diagram, list all required equipment with quantities, estimate costs, and write a security policy document.	teams produce complete network plans with diagrams, equipment lists, and budgets. They present their designs to the class for peer review.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your network design to the class. Explain your topology choice, device selection, and security measures. Answer questions from classmates and the teacher about your design decisions.	students present confidently and defend their choices with reasoning. They receive feedback on completeness, cost-effectiveness, and security of their designs.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Google account access · Cloud storage comparison chart · USB drive sample · Textbook · Projector for demo ·

DIFFERENTIATION **App** Provide partially completed network templates where students fill in equipment details... **On** Students create complete network designs following standard planning guidelines **Adv** Advanced students design multi-building campus networks with redundant connections and... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a simple map of your hostel floor. Spiral: also recall one fact from last week.

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Image Editing Overview

I CAN Understand the difference between raster and vector images · Identify common image file formats and their uses · Recognize the importance of image editing in digital communication ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at this poster on the wall and this photo on the screen. One is made of tiny squares called pixels and the other is made of mathematical lines. What happens when you zoom into each one?	students observe that zooming into the pixel image makes it blurry while the line image stays sharp. They realize images are created in different ways.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open these three image files on the computer. A JPEG photo, a PNG logo, and a SVG icon. Zoom into each one to two hundred percent. Observe what happens to the quality. Note the file sizes of each format.	students discover that JPEG loses quality when zoomed while PNG stays crisp for sharp edges. SVG remains perfect at any zoom level. They notice PNG files are larger.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Raster images like JPEG and PNG are made of a grid of pixels. Each pixel has a color. When you enlarge a raster image the pixels become visible and quality decreases. Vector images like SVG are made of mathematical paths and points. They can be scaled to any size without losing quality. JPEG is best for photos. PNG is best for graphics with transparency. SVG is best for icons and logos.	students create a comparison chart of image formats with columns for type, best use, transparency support, and file size. They understand when to use each format.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Think about all the images you see daily. School posters, textbook illustrations, phone photos, website icons, and social media posts. Classify each as raster or vector and explain which format would be best for creating it.	students categorize images and realize that photos are raster while logos and icons should be vector. They plan to use appropriate formats for their graphics projects.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Answer these questions. What is the difference between raster and vector images? Name two raster formats and two vector formats. When would you choose PNG over JPEG?	students explain pixel versus mathematical composition, name formats correctly, and describe PNG transparency advantage for logos.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Network design templates · Equipment price list · Projector for presentation · Chart paper for diagrams · Textbook reference ·

DIFFERENTIATION **App** Provide pre-made image samples with labels and a simplified format comparison sheet **On** Students complete standard format identification and comparison activities **Adv** Advanced students research image compression algorithms and explain lossy versus lossy... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Open the camera app on your phone and take two photos. Spiral: also recall one fact from last week.

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GIMP Interface

I CAN Navigate the GIMP workspace including menus, toolbox, and panels · Customize the GIMP interface for efficient workflow · Create a new image file with specific dimensions and resolution ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Have you ever used a paint program on a computer? How is it different from painting on paper? What tools would you expect to find in a digital painting program?	students mention undo button, color picker, brushes, and eraser. They realize digital painting offers tools impossible with physical paint.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Launch GIMP on the school computer. Explore the interface without any instructions. Find the menu bar at the top, the toolbox on the left, the layers panel on the right, and the image canvas in the center. Try clicking different tools and see what happens.	students discover GIMP has multiple panels and tools. They find that clicking tools changes the cursor and options bar shows tool settings.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	GIMP or GNU Image Manipulation Program is free image editing software. The interface has the image window in the center, the toolbox on the left with selection tools, paint tools, and transform tools, the menu bar on top with File Edit View and Image menus, and dockable dialogs on the right for layers, channels, and paths. You can rearrange panels by dragging them.	students label a GIMP interface diagram with all major components. They practice opening and closing panels and rearranging the workspace.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Set up your GIMP workspace for the graphics projects we will do this chapter. Create a custom workspace with the toolbox, layers panel, and tool options visible. Save this layout using Windows Dockable Dialogs.	students customize their workspace and learn to reset it if confused. They appreciate having a consistent workspace for efficient editing.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Identify each GIMP interface element. Create a new image with width eight hundred pixels, height six hundred pixels, and resolution three hundred DPI. Save it as a XCF file.	students correctly locate all interface elements and create images with specified settings. They understand XCF is GIMP's native format.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Image samples in different formats · Computer with GIMP installed · Projector · Comparison chart template · Textbook ·

DIFFERENTIATION **App** Provide a GIMP interface map with numbered parts for students to label and match **On** Students complete standard interface exploration and image creation activities **Adv** Advanced students explore GIMP scripting and create simple automated workflows using... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Practice opening GIMP on any available computer tonight. Spiral: also recall one fact from last week.

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LESSON 2.03 · LAB · 40 MINUTES

Selection Tools

I CAN Use rectangular, elliptical, and free-form selection tools · Apply selection modifiers like add, subtract, and intersect · Create precise selections to isolate image elements ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine you want to cut out a circle from a colored paper to paste onto another paper. How would you do it with scissors? Now think about doing the same thing digitally on a photograph. What tool would you need?	students think of scissors and templates for physical cutting. They realize digital selection tools act like virtual scissors for precise cutting.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a photo in GIMP. Use the rectangular selection tool to select a portion of the image. Then use the elliptical tool to select a circular area. Now try the free select tool to trace around an irregular shape. Notice how each tool creates different selection boundaries.	students experience different selection shapes and difficulty levels. They find rectangular is easiest while free select requires careful mouse control.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Selection tools define which part of an image you want to edit. Rectangle select creates square or rectangular selections. Ellipse select creates circular or oval selections. Free select lets you draw any shape by clicking points. The fuzzy select or magic wand selects areas of similar color. Selection modifiers let you add to, subtract from, or intersect with existing selections using the tool options.	students practice each selection tool on sample images. They create complex selections by combining rectangular and elliptical shapes using add and subtract modes.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Using only selection tools, create a selection that isolates just the face of a person in a group photograph. You may need to combine multiple selection tools and use zoom for precision. Practice patience and use Ctrl plus Z to undo mistakes.	students develop precision by combining rectangle select for initial area, ellipse select for head shape, and free select for hair details. They learn zooming helps accuracy.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Complete these tasks. Select a rectangular area and fill it with red color. Select an elliptical area and delete it to create a hole. Combine rectangular and elliptical selections to create a crescent moon shape.	students demonstrate tool mastery by creating accurate selections and combining them creatively. Moon shape requires subtracting circle from rectangle.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Computers with GIMP installed · GIMP interface guide · Projector for demonstration · Practice image files · Textbook ·

DIFFERENTIATION **App** Provide simplified images with clear boundaries for easier selection practice **On** Students complete standard selection exercises using all tool types **Adv** Advanced students use the scissors select tool and foreground select tool for complex... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at objects in your room. Spiral: also recall one fact from last week.

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LESSON 2.04 · LAB · 40 MINUTES

Layers Advanced

I CAN Understand the concept of layers as transparent sheets · Use layer blending modes, opacity, and layer masks · Organize complex projects using layer groups and naming conventions ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about a glass painting where you paint the background on one glass, middle ground on another glass, and foreground on a third glass. When you stack them you see the complete picture. How is this like layers in image editing?	students realize that layers stack like glass sheets allowing independent editing of each element without affecting others.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a GIMP project with multiple layers. Toggle the eye icon next to each layer to hide and show it. Notice how each layer contains different elements. Change the opacity of one layer and see the layers below become visible through it.	students experience how layers work independently. They find that reducing opacity creates transparency effects and hiding layers reveals what is underneath.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Layers are like transparent sheets stacked on top of each other. Each layer can contain different image elements. You can edit one layer without affecting others. Layer opacity controls transparency from fully opaque to fully invisible. Blending modes like multiply, screen, and overlay change how layers interact visually. Layer masks hide or reveal parts of a layer without deleting pixels. Layer groups organize related layers into folders.	students practice creating layers, adjusting opacity, and trying different blending modes. They create a simple composition using at least four layers with proper names.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a composite image with a background landscape, a middle ground with trees, and a foreground with a person. Use separate layers for each element. Apply different blending modes to see creative effects. Use a layer mask on the person layer to blend them naturally into the scene.	students build multi-layer compositions understanding how each element can be adjusted independently. They discover creative blending mode effects.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Describe what layers are and why they are useful. Explain the difference between opacity and blending modes. Create a project with at least five named layers demonstrating different features.	students articulate layer benefits clearly and demonstrate practical skills in creating organized multi-layer projects.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Sample photographs for practice · GIMP installed on computers · Selection tool guide handout · Projector · Textbook ·

DIFFERENTIATION **App** Provide a layer building exercise with pre-made elements that students stack and adjust **On** Students create standard multi-layer compositions following step-by-step instructions **Adv** Advanced students explore layer effects like drop shadows and gradients for... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about your dormitory room. Spiral: also recall one fact from last week.

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Filters and Effects

I CAN Apply blur, sharpen, and artistic filters to images · Understand when and why to use different filter effects · Combine multiple filters to create unique visual effects ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at this blurry photograph. Can you make it sharp again? Now look at this sharp photo. Can you make it look like a painting? What magical tools can transform images in these ways?	students wonder if blur can be reversed and how photos become paintings. They discover filters are powerful transformation tools.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a photograph in GIMP. Go to Filters menu and try Gaussian Blur with radius five. Then try Unsharp Mask to sharpen. Try the Artistic Oilify filter. Try the Distorts iWarp filter. Observe how each filter changes the image differently.	students experience that blur softens details, sharpening enhances edges, oilify creates a painting effect, and iWarp creates 变形 effects. They note some filters have adjustable settings.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Filters apply mathematical transformations to pixel values. Blur filters like Gaussian Blur reduce detail and create soft effects useful for backgrounds. Sharpen filters like Unsharp Mask enhance edges and details useful for improving focus. Artistic filters like Oilify and Charcoal create hand-drawn effects. Distort filters like iWarp and Waves bend and twist image elements. Each filter has settings you can adjust for different intensity levels.	students create a filter reference guide listing each filter type, its effect, common settings, and when to use it. They understand filters are tools not toys.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Take a portrait photograph and apply a sequence of filters to create an oil painting effect. First apply slight blur to reduce noise. Then apply Oilify filter. Then adjust colors using Color Balance. Save each step as a separate layer to show your filter progression.	students learn that combining filters creates more convincing effects than using one filter alone. They practice building effects progressively.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Apply Gaussian Blur to a background layer while keeping the subject sharp. Create a pencil sketch effect using Desaturate and Edge Detect filters. Explain why you chose each filter for its specific purpose.	students demonstrate practical filter knowledge by selecting appropriate filters for specific creative goals and explaining their reasoning.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES GIMP with layer examples · Sample multi-layer projects · Layer blending mode reference · Textbook · Computer access ·

DIFFERENTIATION **App** Provide a filter matching activity with before and after images for students to identify... **On** Students complete standard filter application exercises with specific settings **Adv** Advanced students explore Script-Fu to create custom filter chains for batch processing... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Find a photo on your phone. Spiral: also recall one fact from last week.

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LESSON 2.06 · LAB · 40 MINUTES

Color Correction

I CAN Adjust brightness, contrast, and saturation to improve images · Use color balance to correct color temperature issues · Apply curves and levels for precise tonal adjustments ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at this photo taken in the evening. It looks too dark and yellow. This other photo was taken in shade and looks too blue. How can we fix these color problems to make the photos look natural?	students notice that lighting conditions create color casts. They wonder how to restore natural colors in poorly lit photographs.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a dark photograph in GIMP. Use Colors Brightness-Contrast to increase brightness. Then open a photo with a yellow color cast and use Colors Color Balance to add blue. Try Colors Hue-Saturation to make colors more vibrant. Compare before and after results.	students experience that brightness reveals hidden details, color balance corrects temperature, and saturation makes colors pop. They learn each tool serves a specific purpose.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Color correction fixes problems in photographs. Brightness increases or decreases overall lightness. Contrast increases the difference between light and dark areas. Saturation controls color intensity. Color Balance adjusts the mix of red, green, and blue to correct color casts. Curves gives precise control over tonal range by adjusting specific brightness levels. Levels shows a histogram of tonal distribution and lets you set black and white points.	students practice each adjustment tool on sample images. They create a before and after comparison showing improvement in exposure and color accuracy.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Take three poorly exposed photographs intentionally. One too dark, one too bright, and one with a strong color cast. Use different color correction tools to fix each one. Document your adjustment steps and settings for each correction.	students develop judgment about which corrections to apply and in what order. They learn that sometimes multiple adjustments are needed for the best result.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Correct this overexposed photograph using appropriate tools. Adjust the color balance of this indoor photo to remove the orange cast. Increase the vibrance of this dull landscape without making skin tones unnatural.	students demonstrate understanding of color theory by making corrections that look natural and enhance the image without over-processing.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Sample photographs · GIMP filter menu reference · Filter effect comparison chart · Projector · Textbook ·

DIFFERENTIATION **App** Provide before and after comparison images with sliders showing approximate adjustment... **On** Students complete standard color correction exercises on provided practice images
Adv Advanced students use the Curves tool with the histogram to make precise point-by-point... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Take a photo in your dormitory tonight with the lights on.
Spiral: also recall one fact from last week.

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LESSON 2.07 · LAB · 40 MINUTES

Text and Typography

I CAN Add and format text with fonts, sizes, and colors in GIMP · Understand basic typography principles for readable design · Create text effects using layers and filters ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at these two posters. One uses small plain text and the other uses large colorful creative text. Which one catches your eye first? What makes the second poster more attractive?	students notice that creative text design makes posters more appealing. They identify font choice, size, color, and placement as key design elements.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open GIMP and select the text tool. Click on the canvas and type your name. Try changing the font, size, and color in the tool options. Now create a second text layer with different settings. Try placing text over a photograph and see how it looks.	students experiment with text properties and discover that font choice dramatically changes the message feel. They find that text over images needs careful color choice for readability.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Typography is the art of arranging text for readability and visual appeal. Font choice affects mood. Serif fonts like Times feel formal. Sans-serif fonts like Arial feel modern. Script fonts feel decorative. Font size determines hierarchy. Headlines should be larger than body text. Text color should contrast with background for readability. Letter spacing or kerning affects how text feels. Line spacing or leading affects how comfortable text is to read.	students analyze typography in school posters and textbooks. They create a typography guide listing font choices for different purposes like formal, creative, and fun.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Design a title poster for your class event using GIMP. Choose appropriate fonts for the title and subtitle. Use at least three different text effects like drop shadow, outline, or gradient fill. Ensure the text is readable against the background.	students apply typography principles to create effective poster designs. They learn that text effects should enhance readability not reduce it.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a text composition with at least three different fonts that work together harmoniously. Apply a text effect to make the title stand out. Explain your font choices and why they suit the message.	students demonstrate understanding of font pairing and hierarchy. They justify their choices based on mood and readability principles.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Underexposed photos · Overexposed photos · Color cast examples · GIMP color tools · Histogram reference · Textbook ·

DIFFERENTIATION **App** Provide font pairing suggestions and pre-made text effect templates for practice **On** Students create standard typography exercises following design principles **Adv** Advanced students explore text on path effects and perspective text transformations **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the text on your textbook cover. Spiral: also recall one fact from last week.

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LESSON 2.08 · LAB · 40 MINUTES

Photo Retouching

I CAN Remove blemishes and unwanted objects from photographs · Use clone stamp and healing tools effectively · Retouch portraits while maintaining natural appearance ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at this magazine cover. Do you think the model really looks like that in real life? What tools might have been used to make the skin look so perfect? Is it ethical to change how people look in photos?	students discuss photo retouching in media and debate ethics. They wonder where the line is between enhancement and deception.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a portrait photo in GIMP. Find a small blemish on the face. Use the heal tool to click on the blemish and remove it. Now find an unwanted object in the background. Use the clone stamp tool to paint over it with a nearby texture. Compare before and after.	students experience that heal tool blends texture automatically while clone stamp copies exactly. They practice on small areas first for confidence.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Photo retouching improves images by removing imperfections. The heal tool samples texture from surrounding areas and blends it to cover blemishes. The clone stamp copies pixels from one area to another precisely. The blur tool can soften harsh lines. Color adjustment can improve skin tone. Important principles include keeping skin texture natural, not over-smoothing faces, and preserving the person's actual appearance.	students practice retouching on sample portraits. They learn that good retouching is invisible while bad retouching looks plastic and fake.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Retouch a school group photograph. Remove any distracting background elements using clone stamp. Clean up minor blemishes on faces using heal tool. Adjust brightness so all faces are clearly visible. The goal is to improve the photo while making it look unedited.	students practice subtle retouching that enhances without obvious manipulation. They understand that the best retouching is unnoticeable.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Retouch this portrait to remove three blemishes while keeping the natural skin texture. Remove this unwanted person from the background. Explain the ethical considerations of retouching people in photographs.	students demonstrate technical skill and ethical awareness. They can retouch effectively while discussing when retouching crosses ethical boundaries.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES GIMP text tool · Font samples · Typography poster examples · Google Fonts website · Textbook ·

DIFFERENTIATION **App** Provide photos with obvious blemishes that are easy to identify and remove **On** Students complete standard retouching exercises on provided practice images **Adv** Advanced students explore frequency separation technique for professional portrait... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about photos you have seen on social media. Spiral: also recall one fact from last week.

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Compositing

I CAN Combine elements from multiple images into a single composition · Match color and lighting between different source images · Create seamless blends using layer masks and blending modes ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Have you seen movie posters where actors stand in impossible locations like on top of mountains or in space? How do you think these images are created? What if you could put yourself in any location in the world?	students realize that movie posters combine multiple photographs. They get excited about creating their own composite images.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open two photographs in GIMP. Select an object from the first image and copy it. Paste it into the second image. Notice how the lighting and colors do not match. Try adjusting the pasted layer to blend better with the background.	students discover that combining images requires color and lighting matching. They find that simply pasting looks unnatural without adjustment.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Compositing combines elements from different images into one realistic scene. Steps include selecting and extracting the subject, placing it on a new background, matching color temperature using Color Balance, adjusting brightness and contrast to match the scene, adding shadows to ground the subject, and using layer masks for soft edges. The key to believable compositing is matching light direction, color temperature, and perspective.	students practice compositing by placing a person from one photo onto a different background. They follow a step-by-step workflow ensuring realistic results.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a fantasy composite image placing yourself in a famous location. Extract yourself from a school photo, place yourself on a mountain top or beach, match the lighting and colors, add a realistic shadow, and use a layer mask to blend your feet naturally into the ground.	students create imaginative composites while applying technical skills. They learn that patience and attention to detail create convincing results.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate this composite image for realism. Identify three problems with the lighting or color matching. Create your own composite and have a classmate evaluate it for natural appearance.	students develop critical evaluation skills for compositing quality. They can identify mismatched elements and improve their own work through peer feedback.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Portrait photos for practice · Retouching examples · Ethical guidelines handout · GIMP tools reference · Textbook ·

DIFFERENTIATION **App** Provide pre-extracted subjects with matching backgrounds for simpler compositing practice **On** Students complete standard compositing exercises with provided source images **Adv** Advanced students create complex multi-element composites with atmospheric effects like... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the view from your dormitory window. Spiral: also recall one fact from last week.

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Background Removal

I CAN Remove backgrounds from photographs to isolate subjects · Use selection tools and layer masks for clean extraction · Create transparent backgrounds for design projects ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You see product photos on shopping websites where the item appears on a pure white background. Do you think the photo was taken with a white background or was the original background removed? How would you do this for your own photos?	students realize that many product photos have backgrounds removed and replaced. They want to learn how to isolate objects from their original settings.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a photo of a person against a busy background in GIMP. Try using the fuzzy select tool to select the background. Notice how it selects similar colored areas. Now try the foreground select tool to paint over the person and let GIMP separate them from the background.	students discover that simple backgrounds are easier to remove than complex ones. They find that foreground select tool gives better results for complex shapes.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Background removal isolates the subject from its surroundings. Methods include magic wand for simple uniform backgrounds, foreground select tool for complex subjects, path tool for precise edges, and layer masks for non-destructive editing. After extraction, clean up edges using the eraser with soft brush, refine edges using Select Border, and add the subject to a new background. Save as PNG to preserve transparency.	students practice multiple extraction methods and learn which works best for different situations. They understand that no single method works perfectly for all images.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Remove the background from a school event photograph. Place the extracted subjects onto a plain colored background for a yearbook style photo. Ensure edges are clean and no background remnants remain. Save as PNG for transparency and as JPEG with white background for printing.	students complete a practical background removal project. They learn to output files in appropriate formats for different uses.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Compare three background removal methods on the same image. Rate each for accuracy, speed, and ease of use. Which method gave the best result for hair details?	students critically evaluate different approaches. They find that foreground select tool works best for people while magic wand works for simple objects.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Multiple source photographs · Green screen sample images · Compositing tutorial · Color matching guide · Textbook ·

DIFFERENTIATION **App** Provide photos with solid color backgrounds for easier initial extraction practice **On** Students complete standard background removal exercises with increasing complexity **Adv** Advanced students explore channel-based masking for extracting hair and transparent... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Find a product photo online that has a pure white background. Spiral: also recall one fact from last week.

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LESSON 2.11 · LAB · 40 MINUTES

Poster Design

I CAN Apply design principles to create effective educational posters · Use hierarchy, alignment, and contrast for visual communication · Design posters for school events with clear messaging ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Walk around the school and look at all the posters on the walls. Which poster catches your attention first? What makes it stand out from the others? Think about colors, text size, and images.	students identify that bold colors, large text, and clear images make posters effective. They notice that cluttered posters are harder to read from a distance.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine three school posters on the board. Poster one has too much text in small font. Poster two has a great image but unclear message. Poster three has balanced text, image, and white space. Define what works and what does not in each poster.	students discover that effective posters have clear hierarchy, readable text, and purposeful white space. They learn that less text with bigger font is more effective.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Poster design follows key principles. Hierarchy guides the viewer eye from most to least important elements using size and placement. Contrast makes elements stand out using complementary colors or light versus dark. Alignment creates order by lining up elements on invisible grid lines. Repetition creates consistency by using same fonts and colors throughout. White space gives elements room to breathe and prevents visual clutter.	students analyze poster examples against these principles. They create a design checklist and plan their own poster before opening GIMP.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Design a poster for the annual residential school sports day. Include event name, date, time, venue, and a relevant image. Use hierarchy to make the event name largest, add activities list in medium font, and details in smaller font. Choose colors that are energetic and sporty.	students create complete event posters applying all design principles. They practice planning before designing and iterating based on feedback.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate your poster using the design principles checklist. Is the hierarchy clear? Is there sufficient contrast? Are elements aligned? Is there enough white space? Have a classmate review and suggest improvements.	students critically evaluate their own and peer work. They understand that design is an iterative process of creation and refinement.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Photos with various backgrounds · Background removal examples · New background images · GIMP tools · Textbook ·

DIFFERENTIATION **App** Provide pre-made poster templates where students modify text and images for their... **On** Students create posters from scratch following standard design principles **Adv** Advanced students design poster series with consistent branding across multiple... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at posters in your hostel corridor tonight. Spiral: also recall one fact from last week.

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LESSON 2.12 · LAB · 40 MINUTES

Portfolio Project

I CAN Compile all graphics skills into a cohesive portfolio · Present digital work with proper documentation · Reflect on learning progress and skill development ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Throughout this chapter you have learned selection tools, layers, filters, color correction, text, retouching, compositing, background removal, and poster design. Now you will create a portfolio showcasing your best work in each area.	students 回顾 their progress and feel proud of accumulated skills. They realize a portfolio demonstrates their complete learning journey.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Look at these example portfolios from professional graphic designers. Notice how each portfolio has a consistent style, clear organization, and descriptions explaining the work. What makes a portfolio impressive versus just a collection of files?	students identify that good portfolios tell a story of growth, show variety of skills, and include thoughtful descriptions of each piece.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A portfolio is a curated collection of your best work that demonstrates your skills and growth. Include one example from each lesson topic with a title, description of techniques used, and what you learned. Organize work in logical order from simple to complex. Create a cover page with your name and class. Add an introduction explaining your learning journey. Export as a PDF for easy sharing.	students plan their portfolio structure and select their best work from each lesson. They write descriptions explaining the technical skills demonstrated in each piece.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create your complete graphics portfolio in GIMP. Include a cover page, table of contents, and at least eight pieces of work with descriptions. Ensure consistent formatting throughout. Export as both GIMP XCF files and a PDF portfolio document for sharing.	students compile professional portfolios demonstrating their complete skill set. They practice file management and presentation skills.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your portfolio to the class. Explain your design choices for each piece and what skills you demonstrated. Receive feedback from classmates on both the work quality and presentation effectiveness.	students present confidently and articulate their creative and technical decisions. They learn from peer feedback and recognize their growth.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Poster examples · Design principles handout · GIMP with templates · Color palette reference · Past sports day photos ·

DIFFERENTIATION **App** Provide a portfolio template with pre-made pages where students insert their work and... **On** Students create complete portfolios following standard organization and documentation... **Adv** Advanced students create an online portfolio website using free tools to showcase their... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about which graphics project from this chapter you are most proud of. Spiral: also recall one fact from last week.

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LESSON 3.01 · LAB · 40 MINUTES

What is Multimedia

I CAN Define multimedia and identify its components · Understand how text, audio, video, and graphics work together · Recognize multimedia applications in daily life ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about watching a movie. You see pictures moving, hear music and dialogue, read subtitles, and sometimes interact with the story through games. How many different types of media are working together in this single experience?	students count visual, audio, text, and interactive elements. They realize that movies combine many media forms into one experience.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine these examples on the board. A textbook page uses text and images. A radio program uses audio only. A television show uses video and audio. A website uses text, images, audio, and video. Which of these is multimedia and why?	students identify that websites and television shows are multimedia because they combine multiple media types while radio uses only audio.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Multimedia means multiple forms of media working together. The five elements are text for information, graphics for visual appeal, audio for sound, video for moving images, and animation for motion effects. Multimedia can be linear like a movie that plays from start to finish or interactive like a game where users control the experience. Multimedia applications include education, entertainment, advertising, and communication.	students create a multimedia elements chart with examples of each type. They categorize different media experiences as linear or interactive.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Think about your favorite mobile game or educational app. List all the multimedia elements it uses. How do text, graphics, audio, and video work together to create the experience? Present your analysis to the class.	students analyze apps identifying text labels, graphic characters, sound effects, background music, and animated transitions. They see how elements combine for engagement.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Classify each example as multimedia or single media. Online newspaper. Podcast. Interactive museum kiosk. Television advertisement. Digital textbook with videos.	students correctly identify that interactive kiosk and digital textbook are multimedia while podcast uses only audio. They explain their reasoning.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES All previous lesson files · Portfolio template · PDF export settings · Reflection prompt sheet · Presentation area ·

DIFFERENTIATION **App** Provide picture cards showing different media types for sorting and categorization... **On** Students complete standard multimedia identification and classification exercises **Adv** Advanced students research augmented reality and virtual reality as emerging multimedia... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List all multimedia elements you used today. Spiral: also recall one fact from last week.

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LESSON 3.02 · LAB · 40 MINUTES

Audio Editing Audacity

I CAN Navigate the Audacity interface and understand its main features · Import, record, and export audio files in different formats · Apply basic editing operations like cut, copy, and paste to audio ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Have you ever recorded a voice message on your phone and wished you could remove the um and ah sounds? Or add background music to make it sound professional? What tools would you need for this?	students think about editing voice messages and adding music. They realize audio editing can transform rough recordings into polished content.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Launch Audacity on the school computer. Explore the interface. Find the record button, play button, stop button, and the timeline where audio waves appear. Try recording five seconds of silence and then play it back. Observe the flat line representing silence.	students discover Audacity has transport controls like a tape recorder. They see that silence appears as a flat line and sound appears as wavy lines.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Audacity is free audio editing software. The interface has the transport controls at top left for record, play, stop, and rewind. The timeline shows audio as waveforms from left to right. The track control panel on the left has mute and solo buttons. You can zoom in and out to see detail. Key operations include cut to remove selected audio, copy to duplicate, paste to insert, and trim to keep only selected audio. Export formats include MP3 for compressed sharing and WAV for high quality.	students practice recording and basic editing. They learn to select audio by clicking and dragging, then cut, copy, and paste sections.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Record yourself reading a paragraph from your textbook. Edit out any mistakes or long pauses. Add a two-second silence at the beginning for a professional start. Export the edited audio as an MP3 file with your name in the filename.	students complete a practical editing project. They practice selection precision and learn to maintain audio quality during editing.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Complete these tasks. Record ten seconds of audio. Cut the middle three seconds. Copy and paste the first two seconds to the end. Export as MP3. Explain each step you performed.	students demonstrate competence in basic Audacity operations. They articulate their editing process clearly.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Multimedia examples · Textbook · Computer with internet · Projector · Sample multimedia files ·

DIFFERENTIATION **App** Provide step-by-step picture guides for each Audacity operation with numbered screenshots **On** Students complete standard recording and editing exercises following verbal instructions **Adv** Advanced students explore noise removal and equalization for professional audio quality **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Record a thirty-second voice message describing your day at residential school. Spiral: also recall one fact from last week.

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LESSON 3.03 · LAB · 40 MINUTES

Sound Recording

I CAN Set up microphone and recording environment for quality audio · Understand audio settings like sample rate and bit depth · Record clear voice audio with proper levels and techniques ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about the last audio or video you watched online. Was the voice clear and easy to understand? What made it sound professional versus amateur? What environment and equipment do professional podcasters use?	students recall clear professional audio versus echoey phone recordings. They identify quiet environment and good microphone as key factors.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Test different recording positions with the school microphone. Record the same sentence from ten centimeters, thirty centimeters, and one meter away. Listen to each recording and compare clarity, volume, and background noise.	students discover that distance affects audio quality significantly. Close recordings have more bass and less room echo while distant recordings sound thin and echoey.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Good audio recording requires attention to environment, equipment, and settings. Environment should be quiet with soft surfaces to reduce echo. Position the microphone fifteen to twenty centimeters from your mouth slightly to the side to avoid plosive pops. Sample rate of forty-four point one kilohertz is standard for music. Bit depth of sixteen bits provides good dynamic range. Record at moderate levels to avoid distortion and leave headroom for editing.	students practice proper microphone technique. They learn to check recording levels and adjust gain for optimal sound without clipping.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Set up a recording station in a quiet corner of the classroom. Record a one-minute narration about your favorite place in the school. Focus on speaking clearly, maintaining consistent distance from the microphone, and keeping background noise minimal.	students apply recording techniques in a practical setting. They learn that preparation and environment are as important as equipment.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Record the same paragraph three times with different microphone positions. Identify which recording sounds best and explain why. Rate your recording on clarity, volume consistency, and background noise.	students develop critical listening skills. They can identify quality differences and explain the technical reasons behind them.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Computers with Audacity installed · Microphones · Headphones · Audio sample files · Textbook ·

DIFFERENTIATION **App** Provide a recording checklist with pictures showing correct microphone position and... **On** Students complete standard recording exercises following environmental and technical... **Adv** Advanced students explore stereo recording and binaural audio techniques for immersive... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Find the quietest spot in your hostel. Spiral: also recall one fact from last week.

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LESSON 3.04 · LAB · 40 MINUTES

Audio Effects

I CAN Apply noise reduction to clean up audio recordings · Use equalization to adjust frequency balance · Add effects like reverb and compression for professional sound ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Listen to this recording. It has background hiss, sounds muffled, and the volume goes up and down. Can you imagine using tools to fix each of these problems? What would a professional recording studio do?	students identify three common audio problems and wonder if software can fix them. They learn that effects can dramatically improve audio quality.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a noisy recording in Audacity. Select a quiet section with only background noise. Go to Effect menu and choose Noise Reduction. Click Get Noise Profile. Now select the entire track and apply Noise Reduction. Listen to the improvement. Try different reduction levels.	students experience that noise removal works by identifying background noise pattern and subtracting it from the recording. Too much removal creates artifacts.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Audio effects improve sound quality. Noise reduction removes background hiss and hum by sampling noise and subtracting it. Equalization adjusts bass, mid, and treble frequencies to brighten or warm the sound. Compression reduces the difference between loud and quiet parts for consistent volume. Reverb adds space and depth making recordings sound like they were made in a larger room. Each effect should be used subtly to enhance without distorting.	students practice applying effects to sample recordings. They learn that less is more with audio effects and 过度 processing sounds unnatural.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Take your voice recording from the previous lesson. Apply noise reduction to remove background noise. Use equalization to boost clarity in the voice frequency range. Apply light compression for consistent volume. Export the processed version and compare with the original.	students apply a complete effects chain transforming amateur recordings into professional-sounding audio. They document their settings for each effect.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Apply noise reduction to this recording and explain your settings. Use equalization to make this voice sound warmer. Apply compression to even out the volume levels. Evaluate whether each effect improved or degraded the audio.	students demonstrate practical effects knowledge. They can identify when effects help and when they cause problems.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Microphones · Quiet recording space · Pop filter if available · Audacity software · Recording checklist ·

DIFFERENTIATION **App** Provide pre-configured effect presets with recommended settings for common audio problems **On** Students apply standard effects following step-by-step instructions with specific... **Adv** Advanced students explore multiband compression and dynamic EQ for broadcast-quality... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Listen to a song on your phone and focus on the vocals.
Spiral: also recall one fact from last week.

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LESSON 3.05 · LAB · 40 MINUTES

Video Editing Basics

I CAN Import video clips and organize them in a video editor · Trim, split, and arrange clips on a timeline · Add titles and basic transitions between clips ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Watch this raw footage from a school event. It has long boring parts, shaky camera movements, and random scenes in wrong order. How would you transform this into an exciting three-minute highlight video?	students identify problems in raw footage and brainstorm editing solutions. They realize that editing transforms raw material into compelling stories.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a video editor on the school computer. Import two video clips. Notice how they appear in the media library. Drag them onto the timeline. Play the timeline to see them play in sequence. Try trimming the end of the first clip by dragging its edge.	students experience the basic workflow of importing, arranging, and trimming. They find that the timeline represents the sequence of their video.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Video editing arranges clips to tell a story. Import brings media files into the project. The timeline shows clips from left to right in playback order. Trim removes unwanted parts from the beginning or end of a clip. Split cuts a clip into two parts at a specific point. Transitions like fade and dissolve smooth the change between clips. Titles add text information like names and locations. Always save your project frequently to avoid losing work.	students practice basic editing operations. They learn to use keyboard shortcuts for faster editing and to save frequently.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a one-minute video about a day at residential school. Import at least five video clips. Trim each to keep only the best moments. Arrange them in chronological order. Add a title at the beginning and credits at the end. Add simple fade transitions between clips.	students complete a practical video project. They practice making editorial decisions about what to keep and what to cut.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Trim this thirty-second clip to remove the first five seconds and last three seconds. Split this clip at the point where the action changes. Add a fade transition between these two clips. Explain each editing decision you made.	students demonstrate basic editing competence. They can articulate their editorial reasoning.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Audacity with effects menu · Noisy sample recordings · Effect settings guide · Headphones for critical listening · Textbook ·

DIFFERENTIATION **App** Provide pre-made video clips and a template timeline for simpler editing exercises **On** Students create standard video projects following editing guidelines **Adv** Advanced students explore multi-track editing with separate audio and video tracks **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Watch a short video on YouTube. Spiral: also recall one fact from last week.

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Video Transitions

I CAN Identify different types of video transitions and their uses · Apply transitions appropriately based on content mood · Avoid overuse of transitions that distract from content ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Watch these two video clips. The first cuts abruptly from one scene to the next. The second uses a smooth dissolve to blend one scene into the other. Which feels more natural? When would you use a hard cut versus a soft dissolve?	students notice that abrupt cuts feel jarring while dissolves feel gentle. They begin to understand that transition choice affects emotional response.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open your video project from the last lesson. Try adding different transitions between clips. Use a cut, then a fade to black, then a dissolve, then a wipe. Play each one and note how it changes the feeling of the video.	students experience that each transition creates a different mood. Fade to black feels final. Dissolve feels dreamlike. Wipe feels dynamic.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Transitions connect scenes and control pacing. Cut is instant and most common used for same-time same-place scene changes. Dissolve blends two clips together over one to two seconds for gentle transitions. Fade to black gradually darkens to black screen indicating time passage or scene end. Wipe slides one clip off while another slides on for dynamic transitions. Key rule is to match transition to content. Do not use more than two different transition types in one video.	students practice selecting transitions based on content analysis. They learn that simple is usually better than flashy.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Edit your school day video using only appropriate transitions. Use cuts between same-location scenes. Use dissolves for time passage. Use fade to black at the end. Write a transition guide explaining why you chose each transition at each point.	students demonstrate understanding of transition purpose. They can justify each transition choice with content-based reasoning.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Identify the transition type used between each pair of clips in this sample video. Rate whether each transition is appropriate for the content. Suggest better alternatives where transitions do not match the mood.	students develop critical evaluation skills for transition effectiveness. They can identify mismatches and propose improvements.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Video editing software · Sample video clips · Microphone for narration · Projector for demonstration · Textbook ·

DIFFERENTIATION **App** Provide a transition matching activity where students match transition types to... **On** Students apply standard transitions following editing best practices guidelines **Adv** Advanced students explore custom transitions using masking and keyframe animation **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Watch a movie or TV show tonight and pay attention to transitions between scenes. Spiral: also recall one fact from last week.

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LESSON 3.07 · LAB · 40 MINUTES

Animation Principles

I CAN Understand the twelve principles of animation · Create simple animations using keyframes · Apply timing and spacing for realistic motion ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Watch this bouncing ball animation. Now watch this one. Both show a ball bouncing but one looks realistic and alive while the other looks stiff and robotic. What makes the difference between smooth and jerky animation?	students notice that realistic animation has timing variation and squash-stretch while robotic animation has uniform motion. They want to learn the secrets of smooth animation.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a simple animation tool. Create a ball at the top of the screen. Set a keyframe. Move the ball to the bottom and set another keyframe. Play the animation. Now add a third keyframe where the ball squashes at the bottom. Notice how the motion changes.	students experience that keyframes define important positions and the computer fills in the motion between them. Squash and stretch adds realism.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Animation principles guide believable motion. Squash and stretch gives weight and flexibility. Anticipation prepares the viewer for an action. Staging directs attention to the most important element. Straight ahead action creates fluid natural motion. Follow through and overlapping action show that different parts move at different speeds. Slow in and slow out makes motion feel natural by starting and ending slowly. Arcs follow natural curved paths.	students create a bouncing ball animation applying squash, stretch, and slow in slow out principles. They observe that following principles creates more natural motion.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a simple animation of a character walking across the screen. Apply at least four animation principles. Use anticipation before each step. Add follow through on the arms. Use slow in and slow out for natural movement. Stage the character clearly against the background.	students apply multiple principles to create complex believable motion. They learn that combining principles produces better results than using one alone.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Identify which animation principles are used in this sample animation. Create a bouncing ball animation that demonstrates squash, stretch, and timing. Explain why each principle makes the animation look more realistic.	students demonstrate understanding of animation theory and practical application. They can articulate how principles affect visual quality.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Video editing software · Transition examples · Sample video clips · Transition guide chart · Textbook ·

DIFFERENTIATION **App** Provide pre-made keyframe templates where students adjust timing values to see principle... **On** Students create standard animation exercises following principle guidelines **Adv** Advanced students explore character rigging and walk cycle animation with secondary motion **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Watch a short animation on your phone. Spiral: also recall one fact from last week.

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LESSON 3.08 · LAB · 40 MINUTES

Presentation Software

I CAN Create effective slide presentations with appropriate content density · Use themes, layouts, and master slides for consistent design · Incorporate multimedia elements into presentations ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about the worst presentation you have ever seen. What made it bad? Now think about the best one. What made it engaging and memorable? How would you create a presentation that people actually enjoy watching?	students recall presentations with too much text and boring layouts versus ones with great images and storytelling. They want to learn effective presentation design.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open presentation software and explore the available themes. Notice how each theme changes fonts, colors, and layout. Try the slide master to see how it controls the design of all slides. Create a title slide with the school name and your name.	students discover that themes provide consistent professional design. They learn that slide master controls global design elements.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Effective presentations follow the ten-twenty-thirty rule. Ten slides maximum for a twenty-minute presentation with thirty point minimum font. Each slide should have one main idea. Use images more than text. Use bullet points with maximum six items per slide. Maintain consistent fonts and colors throughout. Use slide transitions sparingly. Include speaker notes for your talking points not on the slide.	students redesign a bad presentation applying these principles. They transform text-heavy slides into visual impactful slides.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a five-slide presentation about your favorite topic. Apply the ten-twenty-thirty rule adapted for five slides. Use consistent theme throughout. Include at least one image per slide. Add speaker notes for each slide explaining what you would say during the presentation.	students create complete professional presentations. They practice balancing visual content with verbal explanation.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate this presentation against the ten-twenty-thirty rule. Identify three slides that have too much text. Suggest image alternatives for text-heavy content. Present your own creation to a partner and get feedback.	students apply evaluation criteria to existing work and receive peer feedback on their own presentations. They iterate to improve.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Animation software · Principles reference chart · Storyboard templates · Sample animations · Textbook ·

DIFFERENTIATION **App** Provide presentation templates with pre-designed slides where students add content to... **On** Students create standard presentations following design principles and guidelines **Adv** Advanced students create interactive presentations with hyperlinks and embedded quizzes **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create three slides about your dormitory on your phone using any presentation app. Spiral: also recall one fact from last week.

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LESSON 3.09 · LAB · 40 MINUTES

Interactive Presentations

I CAN Add hyperlinks and navigation to create non-linear presentations · Embed quizzes and interactive elements for audience engagement · Design presentations that adapt based on audience choices ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	What if a presentation could respond to audience questions and show different content based on what people want to learn? Like a choose your own adventure story but for presentations. How would you build something like this?	students imagine presentations that branch into different paths. They get excited about creating interactive experiences.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a presentation and create two different content paths. Add a button on slide one that links to slide three and another button that links to slide five. Click each button and see how the presentation jumps to different slides.	students discover that hyperlinks can jump to any slide creating branching paths. They realize this enables interactive storytelling.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Interactive presentations break the linear slide-by-slide format. Hyperlinks connect text or objects to specific slides creating navigation options. Action buttons provide clickable areas for audience choices. Embedded quizzes test understanding during the presentation. Branching scenarios let audiences choose their learning path. Tools like Google Slides and PowerPoint support these features. Design interactive elements to be large enough to click and clearly labeled.	students add interactivity to their presentations. They create quiz slides with answer feedback and navigation menus for topic selection.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create an interactive quiz presentation about a chapter from your textbook. Include a title slide, question slides with multiple choice options, correct answer feedback slides, and a final score slide. Each answer choice links to the appropriate feedback.	students build complete interactive quiz experiences. They practice designing clear navigation and meaningful feedback for each answer.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Test your interactive presentation with a classmate. Observe whether they can navigate without instructions. Identify any confusing links or unclear buttons. Revise based on feedback to improve usability.	students test their creations with real users. They learn that good design is invisible and confusing design frustrates users.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Presentation software · Sample presentations · Design principles guide · Image resources · Textbook ·

DIFFERENTIATION **App** Provide a pre-made quiz template where students replace questions and answers with their... **On** Students create standard interactive presentations with branching navigation **Adv** Advanced students create adaptive presentations using macros and conditional logic **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think of a quiz about your school. Spiral: also recall one fact from last week.

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LESSON 3.10 · LAB · 40 MINUTES

Digital Storytelling

I CAN Structure a narrative using beginning, middle, and end · Combine text, images, audio, and video for effective storytelling · Create emotional impact through multimedia story elements ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Tell me about a story that made you cry, laugh, or feel excited. What elements did the storyteller use? Was it the words, the pictures, the music, or the way they told it? How can we use multimedia to create powerful stories?	students recall emotional stories and identify elements that created feelings. They realize multimedia can amplify storytelling power.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Watch three different digital stories about the same topic. One uses only text. One uses text with images. One uses text, images, music, and narration. Rate each for emotional impact and engagement. What made the third version most powerful?	students find that adding multimedia layers increases engagement and emotional connection. They notice music especially affects mood.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Digital storytelling uses multimedia to tell narratives. Structure follows the story arc. Beginning introduces characters and setting. Rising action builds tension and conflict. Climax is the turning point. Falling action resolves the conflict. Conclusion provides closure. Multimedia elements reinforce the narrative. Images show what words describe. Music sets the emotional tone. Narration personalizes the story. Text provides context and captions.	students plan a digital story using the story arc structure. They select multimedia elements that support each story section emotionally.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a two-minute digital story about a personal experience at residential school. Write a script following the story arc. Select or create images that match each story moment. Record narration with appropriate emotional tone. Add background music that enhances the mood. Combine all elements in a video editor.	students produce complete digital stories integrating all multimedia skills. They practice storytelling craft alongside technical skills.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your digital story to the class. After presenting, answer questions about your creative choices. What image was most powerful and why? How did music affect the mood? What would you do differently next time?	students present and defend their creative choices. They receive feedback on both technical execution and storytelling effectiveness.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Presentation software · Quiz question bank · Flowchart templates · Navigation icons · Textbook ·

DIFFERENTIATION **App** Provide story templates with pre-written story arcs where students add personal details... **On** Students create standard digital stories following the story arc structure **Adv** Advanced students create interactive digital stories with audience choices affecting the... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about your first day at residential school. Spiral: also recall one fact from last week.

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LESSON 3.11 · LAB · 40 MINUTES

Podcasting

I CAN Plan and structure a podcast episode with segments · Record and edit multi-segment audio programs · Add intro music, transitions, and outro to podcast episodes ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Do you listen to podcasts? What makes a podcast different from a radio show or an audio recording? What elements do your favorite podcasts include that keep you listening?	students discuss podcast elements like regular episodes, host personality, segments, and production quality. They want to create their own.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Listen to this sample podcast episode. Note the intro music, host introduction, main content, interviews, and outro. Create a timeline showing each segment and its duration. How long is each part?	students discover that podcasts have structured segments with different purposes. They find that intros and outros create professional framing.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A podcast is a regularly released audio program with episodes. Planning includes choosing a topic, writing a script outline, identifying segments, and scheduling release dates. Episode structure typically includes intro music and greeting, segment one with main topic, segment two with interview or discussion, listener questions or feedback, and outro with next episode preview. Recording requires quiet environment and consistent audio levels. Editing removes mistakes and tightens pacing.	students create a podcast planning document with topic, segments, and timing. They write a script outline for a pilot episode.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Record a ten-minute podcast episode about an educational topic. Include a thirty-second intro with music, a main segment of five minutes, a guest interview segment of three minutes recorded with a classmate, and a one-minute outro with music. Edit for clarity and add transitions between segments.	students produce complete podcast episodes. They practice interviewing, segment transitions, and maintaining consistent audio quality throughout.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Listen to your podcast and evaluate it against these criteria. Is the intro engaging? Are segments clearly separated? Is the audio quality consistent? Does the outro encourage listeners to return? What would improve the next episode?	students critically evaluate their own productions. They identify strengths and specific areas for improvement in future episodes.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Storytelling structure guide · Image resources · Music library · Narration recording setup · Video editor · Textbook ·

DIFFERENTIATION **App** Provide podcast script templates with pre-written segments where students add their own... **On** Students create standard podcast episodes following structure guidelines **Adv** Advanced students create podcast series with consistent branding, episode numbering, and... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Listen to a five-minute podcast episode tonight. Spiral: also recall one fact from last week.

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LESSON 3.12 · LAB · 40 MINUTES

Multimedia Project

I CAN Plan and execute a complete multimedia project · Integrate multiple media types into a cohesive final product · Present multimedia work with professional documentation ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You have learned audio editing, video editing, animation, presentations, digital storytelling, and podcasting. Now you will combine all these skills into one comprehensive multimedia project. What topic would you like to showcase using all these media forms?	students brainstorm topics that could be expressed through multiple media. They excitedly plan projects that demonstrate their complete skill set.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine these award-winning student multimedia projects. One is about school heritage using video, audio interviews, and photos. Another is about science experiments using animation, narration, and data visualization. Define what makes each project effective.	students identify that successful projects combine media types purposefully. They notice clear planning, consistent theme, and professional polish.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A multimedia project requires careful planning. Define your topic and audience. Create a project proposal with objectives and media plan. Storyboard each section identifying media type and content. Gather or create all media assets. Build the project in stages. Test with audience and revise. Document your process and choices. Present professionally with introduction and Q and A session.	students create project proposals and storyboards. They plan their work in stages with clear milestones and deadlines.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Execute your multimedia project plan. Create at least four different media elements. Combine them into a final product. Write documentation explaining your creative choices and technical process. Prepare a five-minute presentation about your project for the class.	students produce complete multimedia projects demonstrating integrated skills. They practice project management and documentation.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your multimedia project to the class. Explain your media choices, creative decisions, and technical challenges. Receive feedback from classmates using a structured evaluation form. Reflect on what you would do differently next time.	students present confidently and respond to questions professionally. They learn from peer feedback and self-reflection.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Microphones · Quiet recording space · Audacity software · Royalty-free music · Podcast planning template · Textbook ·

DIFFERENTIATION **App** Provide project templates with pre-defined structure where students fill in content for... **On** Students create complete multimedia projects following standard planning and execution... **Adv** Advanced students create multi-chapter multimedia projects with cross-references and... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a one-page project plan for a multimedia story about your family. Spiral: also recall one fact from last week.

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LESSON 4.01 · CONCEPT · 40 MINUTES

Spreadsheet Overview

I CAN Identify the purpose and benefits of spreadsheet software · Navigate the spreadsheet interface including cells, rows, and columns · Enter and edit data in spreadsheet cells ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine you need to track the marks of all fifty students in your class for five subjects. Doing this on paper would take hours and mistakes would be hard to fix. What if there was a tool that could calculate totals, averages, and grades automatically?	students realize that spreadsheets save time and reduce calculation errors. They imagine automatic grade calculation versus manual addition.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open the spreadsheet application. Click on different cells and notice the cell address in the name box like A1, B2, C3. Type your name in cell A1. Type a number in B1. Try using arrow keys to move between cells. Notice the formula bar shows the cell content.	students discover that cells are identified by column letter and row number. They find that each cell can hold text, numbers, or formulas.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A spreadsheet is a grid of cells organized in rows numbered one, two, three and columns labeled A, B, C. Each cell holds one piece of data. The intersection of column C and row five is cell C5. Spreadsheets are used for data entry, calculations, analysis, and visualization. Key features include formulas that calculate automatically, formatting that makes data readable, sorting that organizes data, and charts that visualize patterns.	students practice navigating and entering data. They create a simple table with student names in column A and marks in column B.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a class attendance tracker for the week. Enter student names in column A. Enter days Monday through Friday in row one. Mark present with P and absent with A. Format the header row with bold text and background color.	students apply spreadsheet basics to create a practical document. They learn formatting improves readability of data tables.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Answer these questions. What is a cell address? How do you move between cells? What are three types of data you can enter in a cell? Create a simple table with five rows and three columns.	students demonstrate understanding of spreadsheet fundamentals. They navigate confidently and enter data correctly.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES All previous lesson tools · Project planning templates · Documentation guidelines · Presentation space · Evaluation forms · Textbook ·

DIFFERENTIATION **App** Provide a pre-made spreadsheet template with labels where students enter their own data... **On** Students create spreadsheets from scratch following verbal instructions **Adv** Advanced students explore named ranges and multiple worksheets within a workbook **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Create a simple spreadsheet on your phone or computer listing five items you need to buy from the store with their estimated prices in a column next to them. Spiral: also recall one...

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LESSON 4.02 · CONCEPT · 40 MINUTES

Data Entry and Formatting

I CAN Enter different data types including text, numbers, and dates · Apply cell formatting for alignment, borders, and colors · Use fill handle and auto-fill for efficient data entry ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at this messy spreadsheet. Numbers are left-aligned, text is right-aligned, there are no borders, and everything looks the same. Now look at this formatted version. Same data but organized with colors, borders, and proper alignment. Which one would you rather use?	students immediately prefer the formatted version. They realize that formatting transforms raw data into readable information.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter a list of numbers in column A. Notice they right-align by default. Enter text in column B. Notice it left-aligns by default. Select all cells with numbers and change alignment to center. Add borders around the data. Apply a background color to the header row.	students discover that spreadsheets automatically detect data types and align accordingly. They learn that formatting makes data presentation professional.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Data types determine how information is stored and displayed. Text aligns left by default. Numbers align right. Dates are stored as serial numbers but displayed as dates. Formatting includes alignment left center right, borders thin thick colored, fill colors for highlighting, font bold italic underline, and number format decimal currency percentage. The fill handle at the bottom-right corner of a selected cell copies patterns to adjacent cells automatically.	students practice formatting a sample data table. They use auto-fill to create number sequences and day lists efficiently.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a formatted monthly expense tracker. Enter expense categories in column A, amounts in column B, and dates in column C. Format amounts as currency with rupee symbol. Format dates consistently. Add borders around all cells. Use fill colors to separate categories. Apply bold formatting to headers.	students create professionally formatted spreadsheets. They practice combining multiple formatting features for clear data presentation.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Format this raw data table with proper alignment, borders, colors, and number formatting. Use auto-fill to extend this number sequence from one to twenty. Explain the difference between formatting cells and changing cell content.	students demonstrate formatting competence. They understand that formatting changes appearance without changing underlying data.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software installed · Practice data sheets · Formatting guide · Textbook · Projector for demonstration ·

DIFFERENTIATION **App** Provide pre-formatted templates where students practice specific formatting tasks on... **On** Students complete standard formatting exercises following guidelines **Adv** Advanced students explore conditional formatting and custom number formats **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the menu in your hostel dining hall. Spiral: also recall one fact from last week.

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LESSON 4.03 · CONCEPT · 40 MINUTES

Basic Formulas

I CAN Write formulas using cell references for automatic calculations · Use arithmetic operators like add, subtract, multiply, and divide · Understand the order of operations in spreadsheet formulas ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	If you change a mark in one cell, should you manually recalculate the total in another cell? What if there were one hundred marks to update? There must be a way for the total to update automatically when you change any individual mark.	students realize that formulas with cell references automatically recalculate when source data changes. This saves enormous time.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter the number ten in cell A1 and twenty in cell B1. In cell C1 type the formula equals A1 plus B1. Press Enter. Now change the value in A1 to fifteen. Watch C1 update automatically. Try other operations using multiply and divide symbols.	students discover that formulas recalculate instantly when referenced cells change. They experience the power of automatic calculation.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Formulas start with equals sign and use arithmetic operators. Plus for addition, minus for subtraction, asterisk for multiplication, slash for division, and caret for exponent. Cell references like A1 refer to the value in that cell not the cell itself. Order of operations follows mathematics. Parentheses first, then exponents, then multiplication and division left to right, then addition and subtraction left to right. Always use cell references not fixed numbers so formulas update automatically.	students practice writing formulas with different operators. They learn to use parentheses to control calculation order.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a simple bill calculator. Enter item names in column A, quantities in column B, and prices in column C. In column D write formulas to multiply quantity by price for each item. In a total cell write a formula to sum all line totals. Change a price and observe the total update.	students apply formulas to real-world calculation scenarios. They see how one change cascades through all dependent formulas.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Write a formula to calculate the area of a rectangle using cell references for length and width. Write a formula to calculate average of five numbers in cells A1 through A5. Explain why cell references are better than fixed numbers in formulas.	students create practical formulas and articulate the benefit of dynamic recalculation over static values.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Raw data samples · Formatting reference sheet · Color accessibility guide · Textbook ·

DIFFERENTIATION	App Provide formula templates with cell references already written where students identify... Advanced students explore absolute cell references with dollar signs for fixed values in...	On Students write standard formulas following operator and reference guidelines SEND Touch and identify network devices with teacher guidance, use simple one-word labels	Adv
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write down the prices of five items you use every day. Spiral: also recall one fact from last week.		

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LESSON 4.04 · CONCEPT · 40 MINUTES

Functions SUM AVERAGE

I CAN Use built-in functions for common calculations · Apply SUM to add ranges of numbers and AVERAGE to find mean · Understand function syntax with arguments and ranges ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Writing a formula to add fifty student marks one cell at a time like equals A1 plus A2 plus A3 would be incredibly long. Is there a shorter way to add an entire column of numbers? Spreadsheet functions make this easy.	students imagine the impossibly long formula and appreciate that functions exist for common calculations like sum and average.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter numbers from one to ten in cells A1 through A10. In cell A11 type equals SUM open parenthesis A1 colon A10 close parenthesis. The result should be fifty-five. Now try equals AVERAGE open parenthesis A1 colon A10 close parenthesis for the mean value.	students experience that functions handle entire ranges with simple syntax. They find SUM adds all values and AVERAGE calculates the arithmetic mean.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Functions are pre-built formulas that perform common calculations. Syntax is function name followed by parentheses containing arguments. SUM adds all numbers in a range. AVERAGE calculates the mean. COUNT counts cells with numbers. MAX finds the highest value. MIN finds the lowest value. Ranges use colon to connect start and end cells like A1 colon A10 means all cells from A1 to A10. Functions can be combined in complex formulas.	students practice each function on sample data. They create a grade sheet using SUM for totals, AVERAGE for class average, and MAX for highest score.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a complete marks sheet for five students in three subjects. Use SUM to calculate each student total. Use AVERAGE to find each student average. Use MAX and MIN to find highest and lowest marks in each subject. Add a class average row using AVERAGE across all students.	students apply multiple functions to create a comprehensive grade analysis. They understand how functions work together for complete data analysis.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Write a SUM function to add cells B2 through B20. Write an AVERAGE function for the same range. Explain what COUNT would return for that range. Use MAX and MIN to find the range of values.	students demonstrate function syntax competence. They can apply appropriate functions for different analytical needs.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Formula reference card · Practice data sheets · Calculator for verification · Textbook ·

DIFFERENTIATION **App** Provide pre-entered data ranges where students practice writing SUM and AVERAGE functions **On** Students apply functions to standard data analysis exercises **Adv** Advanced students explore nested functions like SUMIF and AVERAGEIF for conditional... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Count how many hours you studied this week. Spiral: also recall one fact from last week.

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Charts and Graphs

I CAN Create bar charts, line graphs, and pie charts from spreadsheet data · Choose the appropriate chart type for different data · Customize chart titles, labels, and legends for clarity ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at these two reports about student subject preferences. One shows a table of numbers. The other shows a colorful bar chart with the same data. Which one tells the story faster? What makes charts more powerful than tables for visual communication?	students immediately prefer the chart. They realize charts reveal patterns and comparisons that numbers alone hide.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter subject names in column A and number of students in column B. Select the data and go to Insert menu. Choose Bar Chart. A chart appears. Now try Pie Chart and Line Graph with the same data. Notice how each chart type shows the data differently.	students discover that different chart types emphasize different aspects. Bar charts compare quantities. Pie charts show proportions. Line graphs show trends.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Charts visualize data for better understanding. Bar charts compare quantities across categories using rectangular bars. Pie charts show proportions as slices of a circle where total is one hundred percent. Line graphs show trends over time using connected points. To create a chart select your data, choose the chart type, and customize labels and titles. Good charts have descriptive titles, labeled axes, clear legends, and appropriate colors that do not distract.	students practice creating each chart type. They learn to select data first then choose chart type that best represents their message.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a chart showing the monthly rainfall in Karnataka from June to October. Use a bar chart for monthly comparison. Create a pie chart showing the percentage of rainfall in each month. Add a line graph showing the trend. Customize all three charts with proper titles and labels.	students create multiple chart types from the same data. They practice customizing chart elements for professional presentation.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Choose the best chart type to show student marks distribution. Create a bar chart comparing five students total marks. Add a pie chart showing subject wise mark distribution. Explain why you chose each chart type.	students make informed chart selection decisions. They can justify their choices based on data characteristics and communication goals.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Function reference guide · Practice data sets · Grade sheet template · Textbook ·

DIFFERENTIATION **App** Provide pre-made data sets with suggested chart types for practice **On** Students create standard charts following data visualization guidelines **Adv** Advanced students create combination charts and interactive charts with filters **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about what you eat in the dining hall. Spiral: also recall one fact from last week.

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Sorting Data

I CAN Sort spreadsheet data in ascending and descending order · Sort by multiple columns for complex data organization · Understand how sorting affects row relationships ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Your teacher has a list of one hundred students with names, marks, and ranks. If asked to find the top ten students, how would you do it manually? Would sorting the list by marks make this task instant?	students realize that sorting transforms finding information from time-consuming search to instant visibility. They appreciate the power of ordered data.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter a list of ten students with names and marks in a spreadsheet. Select the marks column. Go to Data menu and choose Sort Smallest to Largest. Observe how the entire rows rearrange keeping names matched with their marks. Now sort Largest to Smallest for ranking.	students discover that sorting rearranges entire rows to maintain data integrity. Each student's name stays with their correct mark.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Sorting rearranges rows based on values in one or more columns. Ascending order puts smallest values first. Descending order puts largest values first. Sort by one column for simple ordering. Sort by multiple columns for complex organization like first by class then by marks within each class. Important rule is to include all columns when sorting so data stays aligned. Never sort only one column as it would separate data relationships.	students practice sorting by different columns and in different orders. They learn to always select the entire data range before sorting.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a student database with columns for name, class, subject, and marks. Sort by marks to find top performers. Sort by class to group students. Sort by class first then by marks within each class for class-wise ranking. Identify the top three students in each class.	students practice multi-level sorting for complex data analysis. They discover that primary and secondary sort keys create meaningful groupings.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Sort this data by student name alphabetically. Then sort by marks from highest to lowest. Create a class-wise sorted list showing top performer in each class. Explain what would happen if you sorted only the marks column without the names.	students demonstrate sorting competence and understand data integrity risks. They can predict problems from incorrect sorting.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Sample data sets · Chart type selection guide · Color printing capability · Textbook ·

DIFFERENTIATION **App** Provide small data sets with clear patterns for easy sorting practice **On** Students complete standard sorting exercises with single and multiple columns **Adv** Advanced students explore custom sort orders and sorting by formula results **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List five classmates with their heights. Spiral: also recall one fact from last week.

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Filtering Data

I CAN Apply filters to display only specific rows matching criteria · Use multiple filter conditions for complex data queries · Clear filters and understand their effect on data view ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You have a spreadsheet with five hundred rows of student data from all residential schools. Your principal asks for only Class Seven students from Bidar district with marks above eighty. How would you find these records quickly without scrolling through everything?	students realize that filtering instantly shows only the records matching their criteria. This saves enormous time compared to manual searching.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter a dataset with columns for name, class, district, and marks. Select the header row. Go to Data menu and enable AutoFilter. Click the dropdown arrow on the class column. Uncheck Select All and check only Seven. Notice only Class Seven rows appear. The row numbers turn blue indicating hidden rows.	students experience that filtering hides non-matching rows rather than deleting them. They see that filtered data is a temporary view of the full dataset.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Filtering hides rows that do not match your criteria showing only relevant data. AutoFilter adds dropdown arrows to headers. Click the arrow to choose filter criteria. Text filters include contains, begins with, equals. Number filters include greater than, less than, between. Multiple column filters combine with AND logic meaning all conditions must be met. Clear Filter shows all rows again. Filters do not delete data they only hide it temporarily.	students practice applying filters to different columns. They learn to combine filters for complex queries like Class Seven AND marks greater than eighty.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a school database with fifty students across three classes and two houses. Filter to show only Class Seven students. Then add a second filter for only Red House students. Now filter for students with marks above seventy-five. Clear each filter one at a time and observe the changes.	students practice layered filtering for complex data queries. They understand how combining filters narrows results progressively.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Apply a filter to show only students from Bengaluru with marks between sixty and eighty. Explain what happens to the row numbers when a filter is active. How would you quickly see all data again after filtering?	students demonstrate filter competence. They understand visual indicators and can reset filters to see complete data.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Practice data sets · Sorting scenarios · Data integrity checklist · Textbook ·

DIFFERENTIATION	App Provide pre-formatted tables with AutoFilter already enabled for simple filter practice On Students apply standard filters following specific query requirements Adv Advanced students explore advanced filter features and database query functions SEND Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about your class. Spiral: also recall one fact from last week.

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LESSON 4.08 · CONCEPT · 40 MINUTES

Conditional Formatting

I CAN Apply color scales to visualize data values with colors · Create data bars that show magnitude visually · Set icon sets to indicate value thresholds ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Look at this marks sheet with five hundred numbers. It is impossible to quickly identify high and low performers. Now look at the same sheet with red for low marks, yellow for average, and green for high marks. How does color transform your ability to understand the data?	students see that conditional formatting makes patterns visible instantly. They can spot high and low values without reading every number.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Enter student marks in a column. Select the marks range. Go to Format menu and choose Conditional Formatting. Apply a color scale from red through yellow to green. Notice how low numbers turn red, middle numbers turn yellow, and high numbers turn green. Try data bars to see length represent value.	students experience that conditional formatting automatically assigns visual properties based on cell values. Colors update when data changes.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Conditional formatting changes cell appearance based on their values. Color scales apply gradient colors from low to high values. Red-yellow-green is common for performance data. Data bars add colored bars proportional to cell value showing magnitude visually. Icon sets add symbols like arrows or traffic lights to indicate value ranges. Rules can use predefined scales or custom criteria like greater than seventy gets green. This makes patterns and outliers immediately visible.	students practice applying different conditional formatting types. They learn to choose appropriate visual indicators for their data.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a class marks sheet and apply three conditional formatting rules. Use color scale for overall mark visualization. Use data bars for comparing student totals. Use icon sets to indicate pass or fail status. Test by changing values to see formatting update automatically.	students apply multiple conditional formatting rules to create informative data visualizations. They see how formatting reveals data stories.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Apply a color scale to this temperature data from red for cold to blue for hot. Create data bars for these sales figures. Add traffic light icons where green is above target, yellow is at target, and red is below target.	students apply conditional formatting appropriate to data context. They demonstrate understanding of when each format type is most effective.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Large practice dataset · Filter scenarios · Data query examples · Textbook ·

DIFFERENTIATION **App** Provide pre-entered data with step-by-step instructions for applying specific... **On** Students apply standard conditional formatting following data visualization guidelines **Adv** Advanced students create custom conditional formatting rules using formulas for complex... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at a traffic light tonight. Spiral: also recall one fact from last week.

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Data Validation

I CAN Set rules that control what data can be entered in cells · Create dropdown lists for consistent data entry · Display input messages and error alerts for data quality ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Your teacher enters student marks but accidentally types one hundred and fifty instead of fifty. The average calculation becomes wrong. How could the spreadsheet prevent this error at the moment of data entry rather than finding it later?	students realize that data validation prevents errors before they happen. This is more efficient than correcting mistakes afterward.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Select a column where marks will be entered. Go to Data menu and choose Data Validation. Set the criteria to whole number between zero and one hundred. Try entering one hundred and fifty. Notice an error message appears preventing the invalid entry. Now try entering eighty and it accepts it.	students experience that data validation rejects invalid entries immediately. They see how validation rules enforce data quality at the source.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Data validation controls what users can enter in cells. Settings include number ranges like zero to one hundred for marks, date ranges for valid date periods, text length limits, and custom formulas for complex rules. Dropdown lists provide predefined options ensuring consistent spelling and terminology. Input messages appear when a cell is selected guiding users. Error alerts appear when invalid data is entered with options to retry or cancel. This prevents data quality problems before they occur.	students practice setting different validation rules. They create dropdown lists for class selection and number ranges for marks.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a student data entry form with validation. Column A for names with text length maximum fifty. Column B for class with dropdown list of classes. Column C for marks with number between zero and one hundred. Column D for grade with dropdown list. Test each validation by trying invalid entries.	students build complete validated data entry systems. They understand that validation creates reliable data for analysis.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Set up validation to accept only dates in the current school year. Create a dropdown list with five subject names. Explain what happens when a user tries to enter invalid data. How would you help a user who sees an error message?	students demonstrate validation setup competence. They can guide users through validation error resolution.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Sample data sets · Conditional formatting reference · Color scale examples · Textbook ·

DIFFERENTIATION	App Provide templates with pre-set validation rules where students test and document the... students create dependent dropdown lists where options change based on previous...	On Students create standard validation rules following data quality guidelines SEND Touch and identify network devices with teacher guidance, use simple one-word labels	Adv Advanced
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about forms you fill out online. Spiral: also recall one fact from last week.		

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LESSON 4.10 · CONCEPT · 40 MINUTES

Pivot Tables Intro

I CAN Understand what pivot tables do and when to use them · Create a basic pivot table from a data set · Summarize data using rows columns and values ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You have a spreadsheet with one thousand rows of sales data from all residential school stores. The principal wants to know total sales per school and average sale per item. Creating formulas for each school would take days. Is there a tool that can summarize large data instantly?	students realize that pivot tables can summarize thousands of rows into useful summaries in seconds. They want to learn this powerful tool.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a sample sales dataset with columns for school, item, quantity, and price. Select all data. Go to Insert menu and choose Pivot Table. A dialog appears asking where to place it. Choose a new worksheet. Now drag school to Rows area and price to Values area. Watch the summary appear instantly.	students experience that pivot tables automatically group and summarize data. They see that dragging fields to different areas changes the summary view.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Pivot tables summarize large datasets into meaningful reports. They have four areas. Rows define what to group by like school names. Columns define what to subdivide by like item types. Values define what to calculate like sum of price. Filters define what subset to include. You can drag fields between areas to change the summary. Double-click values to see the detail behind any summary number. Pivot tables update when source data changes.	students practice creating pivot tables with different field arrangements. They learn to read and interpret pivot table summaries.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a pivot table from student marks data. Put class in Rows and subject in Columns and marks in Values using average function. This creates a class-wise subject-wise average marks table. Add a filter for house to see averages for specific houses only.	students create multi-dimensional summaries using pivot tables. They discover how quickly pivot tables answer complex data questions.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a pivot table showing total marks per student. Create another showing average marks per subject. Explain what each summary tells the principal that raw data does not.	students demonstrate pivot table creation and interpretation. They can explain the analytical value of summarized data.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Validation examples · Dropdown list templates · Error message examples · Textbook ·

DIFFERENTIATION **App** Provide pre-built pivot tables where students practice reading and interpreting the... **On** Students create basic pivot tables following standard field placement guidelines **Adv** Advanced students explore calculated fields and pivot charts for dynamic reporting **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Count how many hours you spent on each subject this week. Spiral: also recall one fact from last week.

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Data Analysis

I CAN Use multiple spreadsheet tools together for comprehensive analysis · Interpret data patterns and draw meaningful conclusions · Present data findings with charts and written analysis ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Your school wants to improve student performance. You have marks data from the last three exams. How would you analyze this data to find which subjects need more attention, which students are improving, and which are falling behind?	students realize that raw data needs analysis to become actionable insights. They plan to use all the spreadsheet skills they have learned.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a comprehensive student marks dataset. Apply sorting to rank students. Use filters to focus on specific classes. Calculate averages with functions. Create charts to visualize trends. Apply conditional formatting to highlight concerns. Notice how each tool reveals different insights.	students experience that combining multiple analysis tools provides complete understanding. Sorting shows rankings, filtering isolates groups, functions calculate metrics, and charts reveal patterns.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Data analysis combines multiple techniques. First organize data with proper formatting and validation. Then sort to see rankings. Filter to focus on specific groups. Calculate key metrics using SUM, AVERAGE, MAX, MIN. Visualize patterns with appropriate charts. Apply conditional formatting to highlight important values. Finally interpret findings and write conclusions. The goal is to transform raw data into actionable insights for decision making.	students follow a structured analysis workflow on sample data. They learn that systematic analysis produces reliable insights.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Define three-exam marks data for your class. Sort students by improvement between exam one and exam three. Create charts showing subject-wise class averages. Identify the three students with most improvement and three with most decline. Write a one-page analysis report with recommendations.	students complete comprehensive analysis projects. They practice data interpretation and report writing for real-world application.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your data analysis findings to the class. Explain what tools you used and why. Describe three key findings from your analysis. Recommend two actions the school could take based on your findings.	students present data-driven recommendations. They demonstrate ability to translate data patterns into practical suggestions.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Large sample dataset · Pivot table field list guide · Summary examples · Textbook ·

DIFFERENTIATION **App** Provide structured analysis worksheets with guiding questions for each analysis step **On** Students complete standard analysis following the organized workflow **Adv** Advanced students explore statistical functions like standard deviation for deeper... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about your marks in the last three tests. Spiral: also recall one fact from last week.

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Budget Project

I CAN Create a complete budget spreadsheet with income and expenses · Apply formulas for totals, balances, and percentage calculations · Use charts to visualize budget allocation and spending patterns ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine you receive five thousand rupees pocket money each month from your family at residential school. You need to manage food, supplies, entertainment, and savings. How would you track every rupee and ensure you do not run out of money before the month ends?	students realize that budgeting requires careful tracking and planning. They want to create a system that helps them manage money wisely.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open a blank spreadsheet. Create columns for Date, Description, Category, Amount In, and Amount Out. Enter some sample income and expense entries. In a summary area use SUM to calculate total income and total expenses. Calculate the balance as income minus expenses.	students experience creating a basic budget structure. They see how formulas automatically calculate running totals.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A budget tracks income and expenses over a period. Structure includes income section listing all money received, expense section listing all money spent, category column for organizing expenses like food, transport, supplies, entertainment, savings, and summary section with totals and balance. Formulas needed include SUM for totals, subtraction for balance, percentage formulas for category spending as percent of total. Charts help visualize where money goes. Update the budget daily for accuracy.	students build complete budget spreadsheets. They practice entering realistic residential school expenses and calculating meaningful summaries.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a monthly budget for managing five thousand rupees at residential school. Categorize expenses into food, stationery, transport, entertainment, and savings. Enter daily transactions for one simulated week. Use formulas to calculate category totals and percentage of total spending. Create a pie chart showing expense distribution. Write recommendations for reducing unnecessary spending.	students create functional budget tools. They practice realistic financial planning and data visualization for personal finance management.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your budget spreadsheet. Show the total income, total expenses, and remaining balance. Create a chart showing spending by category. Explain two ways the budget could be improved for next month.	students demonstrate budget creation and financial analysis skills. They can identify spending patterns and suggest improvements.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Comprehensive student dataset · Analysis template · Chart examples · Report writing guide · Textbook ·

DIFFERENTIATION	App Provide budget templates with categories and formulas already set up where students... Advanced students create annual budgets with monthly tracking and trend analysis	On Students create complete budget spreadsheets from scratch following guidelines	Adv Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Tonight write down everything you spent money on today. Spiral: also recall one fact from last week.		

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LESSON 5.01 · DEBUG · 40 MINUTES

Scratch Review

I CAN Recall Scratch interface elements and basic block categories · Review event blocks, motion blocks, and control blocks · Rebuild a simple project from memory to reinforce fundamentals ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think back to when you first used Scratch. What were the first three blocks you ever used? Can you remember what they did? Today we will refresh our memory and prepare for more advanced programming.	Students recall green flag event, move steps block, and say hello block. They feel nostalgic and ready to build on their foundation.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch without any instructions. Try to recreate your very first project from last year. Use whatever blocks you remember. If you get stuck, look at the block palette and see if any blocks look familiar.	Students struggle with some forgotten concepts but recover quickly as they see familiar blocks. They realize they remember more than they thought.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Scratch has a stage where the program runs, sprites that perform actions, and scripts that control sprite behavior. Block categories include Events for starting scripts, Motion for movement, Looks for appearance, Sound for audio, Control for loops and conditions, Sensing for interaction, Operators for math and logic, and Variables for storing data. Each block snaps together like puzzle pieces to build programs.	Students create a reference sheet listing all block categories with two example blocks from each. They rebuild a simple animation from memory.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Recreate a project that makes a character walk across the screen, say hello, and change color. Use at least five different block categories. Add a second sprite with its own script. Make both sprites respond to the green flag click.	Students rebuild projects using accumulated knowledge. They practice combining blocks from different categories for richer interactions.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	List all eight block categories with one block from each. Explain what the green flag does. Describe the difference between a forever loop and a repeat loop. Debug this broken script and explain what was wrong.	Students demonstrate fundamental Scratch knowledge. They can identify, explain, and fix basic program elements.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES

DIFFERENTIATION **App** Provide block matching cards where students connect block pictures to their functions **On** Students complete standard review exercises rebuilding basic projects **Adv** Advanced students review advanced blocks like clones and custom blocks for upcoming... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Open Scratch on any available device tonight and try to make a cat sprite say your name. Spiral: also recall one fact from last week.

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LESSON 5.02 · DEBUG · 40 MINUTES

Variables Advanced

I CAN Create and use variables for storing and updating values · Understand local versus global variable scope · Apply variables in calculations and program logic ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine a game where your score increases every time you catch a coin. Where does the game store the current score? How does it remember the score even after the coin disappears? What programming tool handles this memory?	Students realize that variables act as memory storage for programs. They understand that games need to remember scores, lives, and levels.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and create a new variable called score. Notice it appears in the variables palette and optionally on the stage. Use set score to zero block. Then use change score by one block. Click these blocks and watch the score variable update on the stage.	Students experience that variables hold values that can be changed. They see the variable value update in real-time on the stage.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Variables are named containers that store data values. Create a variable with the Make a Variable button. Set block assigns an initial value. Change block adds to the current value. Variables can hold numbers for scores and levels, text for names and messages, or true or false for game states. For all sprites scope means every sprite can read and change the variable. For this sprite only scope means only that sprite can access the variable.	Students create variables for different purposes. They practice using variables in calculations and understand scope differences.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a counter program. Make a variable called count. Set it to zero. Create two buttons using sprites. One button increases count by one when clicked. The other decreases count by one when clicked. Display the count value on the stage. Add a reset button that sets count back to zero.	Students build practical variable applications. They learn that variables enable user interaction and dynamic program state.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a variable called timer that counts down from sixty. Use the change and set blocks correctly. Explain when you would use for all sprites versus for this sprite only scope.	Students demonstrate variable creation and scope understanding. They apply variables to practical timing scenarios.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Spreadsheet software · Budget template · our school expense examples · Chart tools · Financial literacy guide · Textbook ·

DIFFERENTIATION **App** Provide variable creation exercises with step-by-step instructions and visual block guides **On** Students create standard variable applications following programming guidelines **Adv** Advanced students explore lists as collections of variables and create scoreboards with... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write down three things a video game needs to remember while playing. Spiral: also recall one fact from last week.

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LESSON 5.03 · DEBUG · 40 MINUTES

Conditions If-Else

I CAN Use if-then blocks to make decisions in programs · Apply if-then-else blocks for binary choices · Combine conditions with comparison and logical operators ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about getting dressed in the morning. If it is raining you take an umbrella. If it is sunny you wear sunglasses. Your brain makes decisions based on conditions. How do we teach a computer to make similar decisions?	Students relate programming conditions to daily decisions. They realize that computers need explicit rules to make choices.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and create a simple program. Make the sprite move ten steps. Now wrap it in an if-then block with the condition touching edge. Inside the if put a turn fifteen degrees block. Run it and watch what happens when the sprite reaches the edge.	Students see that the sprite only turns when the condition is true. They experience conditional execution firsthand.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	If-then blocks check a condition and execute the enclosed blocks only if the condition is true. If-then-else blocks provide two paths. If true the first set of blocks runs. If false the second set runs. Comparison operators include equals, greater than, less than. Logical operators include and for both conditions true, or for either condition true, and not for reversing a condition. Conditions can check sprite positions, variable values, key presses, and other states.	Students practice creating programs with decisions. They learn to choose appropriate conditions and operators for different scenarios.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a quiz program. Ask a question using ask block. Use if-then-else to check if the answer equals the correct response. If correct increase score by one and say correct for two seconds. If wrong say try again and decrease lives by one.	Students build interactive programs with conditional logic. They experience how conditions create branching program behavior.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Write a program that makes a sprite grow big if the space key is pressed and shrink if the down arrow is pressed. Explain the difference between if-then and if-then-else blocks. Create a condition that checks if score is greater than ten.	Students demonstrate condition competence. They understand when each block type is appropriate.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software or website · Previous year projects · Block reference guide · Debugging examples · Textbook ·

DIFFERENTIATION **App** Provide pre-written conditions where students identify whether each is true or false for... **On** Students create standard if-then and if-then-else programs **Adv** Advanced students explore nested conditions and compound logical operators for complex... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write three if-then sentences about your morning routine.
Spiral: also recall one fact from last week.

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LESSON 5.04 · DEBUG · 40 MINUTES

Nested Conditions

I CAN Create programs with conditions inside other conditions · Understand the order of evaluation in nested if blocks · Apply nested conditions for complex decision scenarios ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about ordering food at a restaurant. First you decide if you want vegetarian or non-vegetarian. Then within that choice you decide if you want spicy or mild. You make a decision inside another decision. How would you program this layered decision making?	Students see that real decisions often happen in layers. They realize nested conditions model complex real-world choices.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and create a program with an if-then block that checks if the mouse is down. Inside that if put another if-then that checks if the mouse is touching a specific sprite. Notice that the inner condition only runs when the outer condition is already true.	Students experience that nested conditions create gates. The inner condition cannot be checked unless the outer condition passes first.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Nested conditions place one if block inside another. The outer condition is checked first. If it is false the inner condition is never checked. If it is true the inner condition is then evaluated. This creates a hierarchy of decisions. The program flows from outer to inner conditions. Each level narrows the requirements. Nested conditions can be three or four levels deep but too many levels make code hard to read. Alternatives include combining conditions with logical operators.	Students practice building nested condition programs. They learn to choose between nesting and combining conditions based on clarity.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a character customization program. First check if a costume button is clicked. Inside that check which character type is selected. Inside that check which color is chosen. Apply the complete selection to the sprite. Make sure each nesting level has a clear purpose.	Students build multi-level decision programs. They practice organizing complex logic in readable nested structures.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Trace through this nested condition program. What happens when the outer condition is true and the inner is false? Create a program with three nesting levels. Explain why nesting was better than combining all conditions in one if block for your example.	Students can trace program execution through nested conditions. They justify their design choices with reasoning.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · Variable reference sheet · Counter activity guide · Scope diagram · Textbook ·

DIFFERENTIATION **App** Provide partially completed nested condition programs where students fill in the missing... **On** Students create standard nested condition programs following guidelines **Adv** Advanced students explore deeply nested conditions and refactor them into cleaner... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about choosing what to eat. Spiral: also recall one fact from last week.

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Loops Advanced

I CAN Use repeat-until loops for condition-based repetition · Combine loops with conditions for complex behavior · Avoid infinite loops by ensuring termination conditions ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about washing dishes. You keep washing until all dishes are clean. You do not count how many dishes there are. You stop when a condition is met. How would you program a computer to repeat until a condition changes rather than repeating a fixed number of times?	Students realize that repeat-until loops are more flexible than repeat loops. They understand condition-based repetition matches many real activities.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and create a repeat-until loop that makes a sprite move until it touches the edge. Use the touching edge condition. Add a turn block inside the loop. Watch the sprite move and turn until it reaches the boundary.	Students see that repeat-until keeps executing while the condition is false and stops when it becomes true. They observe the sprite behavior change dynamically.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Repeat-until loops execute blocks repeatedly until a condition becomes true. The condition is checked before each iteration. When the condition is true the loop stops. This differs from repeat which runs a fixed number of times. Repeat-until is better when you do not know how many iterations are needed. Important safety rule always ensure the condition will eventually become true to avoid infinite loops. Infinite loops freeze the program and require force quitting.	Students practice creating repeat-until loops with various conditions. They learn to verify that their termination conditions will eventually be met.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a maze game. The sprite starts at the beginning and the player uses arrow keys to navigate. Use a repeat-until loop that runs until the sprite touches the goal. Inside the loop check key presses and move accordingly. Add a win message when the goal is reached.	Students build interactive games using repeat-until loops. They experience how loops create continuous gameplay that ends at a specific condition.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a repeat-until loop that counts down from sixty to zero. Explain why this loop will not run forever. Create a loop that stops when two conditions are both true. Identify the termination condition in each program.	Students demonstrate understanding of loop termination. They can verify that their loops will end correctly.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · Condition examples · Flowchart templates · Comparison operator reference · Textbook ·

DIFFERENTIATION **App** Provide loop completion exercises where students predict how many iterations will occur... **On** Students create standard repeat-until programs with verified termination conditions
Adv Advanced students combine repeat-until with nested conditions for complex interactive... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about a task you repeat every day until it is done.
Spiral: also recall one fact from last week.

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LESSON 5.06 · DEBUG · 40 MINUTES

Lists

I CAN Create and manipulate lists to store multiple values · Use list operations like add, delete, and insert · Apply loops to process all items in a list ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about your class attendance register. It stores fifty student names in a specific order. You can add new names, remove names, and look up any name by position. How would you create this kind of ordered collection in a program?	Students realize that variables store single values but lists store many values. They see lists as ordered collections like registers and inventories.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and create a new list called names. Add your name using add thing to names block. Add three classmates names. Use delete block to remove the second item. Use insert block to add a name at position two. Look at how the list changes on the stage.	Students experience list manipulation operations. They see that lists display on the stage and can be modified dynamically.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Lists store ordered collections of values. Create a list with Make a List button. Add block appends an item to the end. Delete block removes an item by position. Insert block adds an item at a specific position. Replace block changes an item at a position. Item of block reads a value by position. Length of block counts items. Contents of block reports all items. Lists use positions starting from one not zero. Loops can process every item in a list using the length to determine iterations.	Students practice all list operations. They learn to read, write, and modify list contents programmatically.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a to-do list application. Make a list called tasks. Add buttons to add new tasks, mark tasks complete by deleting them, and display all tasks. Use a loop to number each task when displaying. Count the remaining tasks using length.	Students build practical list-based applications. They experience how lists enable dynamic data management in programs.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a list of five favorite foods. Delete the third item. Insert a new item at position one. Replace the last item. Write a loop that says each item. Explain what happens if you try to access item ten in a list with only five items.	Students demonstrate list manipulation competence. They understand list boundaries and error handling.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · Nested condition examples · Decision tree templates · Tracing worksheets · Textbook ·

DIFFERENTIATION **App** Provide pre-made lists where students practice reading and modifying with guided... **On** Students create and manipulate lists following standard operations **Adv** Advanced students explore parallel lists and create databases with multiple related lists **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a list of five things you need to pack when going home for holidays. Spiral: also recall one fact from last week.

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LESSON 5.07 · DEBUG · 40 MINUTES

Clones

I CAN Create clones of sprites for dynamic program elements · Use clone-specific blocks for individual clone behavior · Manage clone lifecycle from creation to deletion ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about a game where enemies appear from the right side of the screen repeatedly. Do you create a hundred separate sprites or do you create one enemy sprite that copies itself? How would a computer handle making multiple copies of the same character?	Students realize that creating hundreds of sprites would be inefficient. They wonder how games create many instances of the same enemy efficiently.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and create a coin sprite. Add a green flag event and a forever loop. Inside the loop add create clone of myself block and wait one second. Now add when I start as clone block with a go to random position and fall down script. Watch clones appear and fall.	Students experience that clones behave like copies of the original sprite. They see that each clone runs its own when I start as clone script.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Clones are temporary copies of a sprite created during program execution. Use create clone of block to make a new clone. The clone starts running scripts under when I start as clone hat block. Each clone is independent and can have different position and costume. Clones can create more clones but Scratch limits total clones to three hundred. Use delete this clone block to remove clones when they are no longer needed. Clones are destroyed when the program stops.	Students practice creating and managing clones. They learn to control clone creation rate and deletion for efficient programs.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a rain simulation. Make a raindrop sprite that clones itself repeatedly. Each clone starts at the top of the screen in a random horizontal position and falls to the bottom. When a clone reaches the bottom it is deleted. Add slight wind effect by moving clones sideways as they fall.	Students build particle simulation effects using clones. They experience how clones create complex visual effects efficiently.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a bullet system where a spaceship sprite clones a bullet when space is pressed. Each bullet moves upward and is deleted when it reaches the top. Explain why clones are better than creating individual bullet sprites. How would you limit the number of clones on screen?	Students demonstrate clone management competence. They understand memory implications and practical limitations.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · Loop examples · Infinite loop warning · Termination condition guide · Textbook ·

DIFFERENTIATION **App** Provide clone creation exercises with visual guides showing sprite to clone relationships **On** Students create standard clone applications following guidelines **Adv** Advanced students explore clone communication using broadcasts for coordinated clone... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about a game you play. Spiral: also recall one fact from last week.

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LESSON 5.08 · DEBUG · 40 MINUTES

Pen Extensions

I CAN Use the pen extension to draw graphics programmatically · Control pen color, size, and transparency for visual effects · Create geometric patterns and art using loop and pen blocks ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Watch this animation where a sprite draws a perfect square as it moves. Now it draws a spiral. Now a flower pattern. The sprite is creating art using only movement commands. How can moving and turning create beautiful drawings?	Students are amazed that simple movement can create complex art. They want to learn the drawing commands.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and add the pen extension from the extensions button. Create a green flag event. Add erase all to clear previous drawings. Add pen down to start drawing. Make the sprite move one hundred steps and turn ninety degrees four times. A square appears.	Students discover that pen down activates drawing as the sprite moves. They see that loops create geometric shapes automatically.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	The pen extension lets sprites draw on the stage. Pen down starts drawing a trail as the sprite moves. Pen up stops drawing. Set pen color changes the drawing color. Set pen size changes line thickness. Erase all clears the stage. Change pen color by number creates rainbow effects. Combine pen blocks with motion and loops to create geometric art. Repeat loops with turn angles create polygons and flowers. Nested loops create complex recursive patterns.	Students practice creating geometric shapes with pen commands. They learn to calculate turn angles for different polygons like three hundred sixty divided by number of sides.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a spirograph art generator. Use nested loops where the outer loop turns a small angle and the inner loop draws a circle. Change pen color in each iteration for rainbow effect. Experiment with different angle combinations to discover unique patterns.	Students create complex generative art using nested loops and pen commands. They discover mathematical beauty in programming.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Draw a regular hexagon using pen and motion blocks. Create a pattern with eight triangles rotated around a center point. Explain why the turn angle must be exactly three hundred sixty divided by the number of sides for regular polygons.	Students demonstrate geometric drawing competence. They connect programming to mathematical concepts.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · List operations reference · To-do list activity guide · Array concept diagram · Textbook ·

DIFFERENTIATION **App** Provide pre-written pen programs where students modify parameters to change the output **On** Students create standard geometric shapes following mathematical guidelines **Adv** Advanced students explore fractal patterns and recursive drawing algorithms **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Draw a simple geometric pattern on paper using a ruler and protractor. Spiral: also recall one fact from last week.

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LESSON 5.09 · DEBUG · 40 MINUTES

Music Extension

I CAN Add musical notes and rhythms to Scratch projects · Create melodies by sequencing note blocks · Combine music with animation for multimedia projects ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about your favorite song. It is made of individual notes played in a specific rhythm. Can you program a computer to play a melody you know? What blocks would you need to control pitch and timing?	Students want to recreate familiar melodies. They realize that music is a sequence of notes with specific durations.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open Scratch and add the music extension. Play a middle C note for one beat. Then play D for one beat. Then E. Then F. Then G. You just played the first five notes of a scale. Try changing note values and durations to create different melodies.	Students discover that music blocks control pitch with note numbers and duration with beats. They experiment with creating simple melodies.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	The music extension provides blocks for creating audio. Play note block specifies pitch using note numbers where sixty is middle C. Play note for beats controls duration where one beat is one quarter note at the current tempo. Set tempo controls speed in beats per minute. Drum blocks add percussion sounds. Rest for beats creates silence between notes. Combine music blocks with say blocks to display lyrics while music plays. Loops can repeat musical phrases.	Students practice creating melodies by sequencing note blocks. They learn to read simple musical notation and translate it to Scratch blocks.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Program a sprite to play the residential school anthem or a familiar folk song. Write out the note sequence on paper first. Translate note letters to Scratch note numbers. Add appropriate timing with beat durations. Sync animation with the music by adding costume changes on beat.	Students create complete musical performances combining audio and visual elements. They practice planning before programming.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Create a program that plays a five-note ascending scale. Add a second program that plays a descending scale. Combine both into a complete up-and-down scale. Explain how changing the tempo value affects the music playback speed.	Students demonstrate musical programming competence. They understand the relationship between tempo and playback speed.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · Clone examples · Rain simulation template · Object concept diagram · Textbook ·

DIFFERENTIATION **App** Provide pre-written note sequences where students arrange them to create simple melodies **On** Students create standard melodies following musical notation guidelines **Adv** Advanced students explore harmony by playing multiple notes simultaneously and create... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Hum a song you know and try to identify the notes. Spiral: also recall one fact from last week.

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Game Mechanics

I CAN Design core game mechanics including scoring and lives · Implement collision detection for interactive gameplay · Create game loops that manage start, play, and end states ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about your favorite game. What makes it fun? Is it the challenge of avoiding obstacles, the satisfaction of collecting items, or the excitement of reaching new levels? What rules govern how the game works?	Students analyze game elements and identify mechanics like scoring, collision, and level progression. They want to create games with similar engaging mechanics.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Play this simple Scratch game. Notice how the score increases when collecting coins. Lives decrease when touching enemies. The game ends when lives reach zero. Identify the three core mechanics: scoring, health, and game over condition.	Students experience game mechanics from a player perspective. They identify how variables, conditions, and loops create the game experience.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Game mechanics are rules that control gameplay. Scoring uses a variable that changes based on player actions. Lives or health uses a variable that decreases when the player fails. Collision detection uses touching blocks to check if sprites interact. Game states include start screen, active gameplay, pause, and game over. The game loop uses a forever loop that checks conditions and updates the game each frame. Balancing mechanics means making the game challenging but not impossible.	Students design game mechanics on paper before programming. They plan scoring rules, life systems, and win conditions.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a simple collection game. The player moves a character with arrow keys. Coins spawn randomly and the player must collect them. Each coin adds ten to score. An enemy moves back and forth and touching it costs one life. Three lives total and game ends when all lives are lost.	Students implement complete game mechanics. They practice combining movement, collision, scoring, and game state management.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Test your game with a classmate. Observe whether the difficulty feels balanced. Is scoring too easy or too hard? Are lives fair? Adjust your game parameters and explain each change you made to improve the experience.	Students learn game testing and balancing. They understand that good game design requires iteration based on player feedback.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software with pen extension · Geometry reference · Pattern examples · Angle calculation guide · Textbook ·

DIFFERENTIATION	App Provide game templates with pre-built mechanics where students modify parameters and add... students create power-up systems, multiple levels, and adaptive difficulty...	On Students create standard games following design guidelines	Adv Advanced students create power-up systems, multiple levels, and adaptive difficulty...	SEND Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Play a mobile game for five minutes tonight. Spiral: also recall one fact from last week.			

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LESSON 5.11 · DEBUG · 40 MINUTES

AI and ML Intro

I CAN Understand the basic concepts of artificial intelligence and machine learning · Explore how AI recognizes patterns in data · Create a simple Scratch project that simulates AI decision making ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Think about how your phone recognizes your face to unlock. How does it know it is you and not someone else? Has the phone met you before? How did it learn to recognize your face among billions of faces?	Students wonder how devices learn to recognize individuals. They realize AI must learn from examples rather than being explicitly programmed for every possibility.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Look at the Scratch ML extensions. Try the text recognition extension by typing hello and seeing how Scratch classifies it. Try the face detection extension to see how it tracks face position. Notice that Scratch can identify objects and detect motion.	Students experience machine learning through Scratch extensions. They see that AI can recognize text, faces, and objects from training data.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Artificial intelligence or AI is computer software that mimics human thinking. Machine learning or ML is a type of AI that learns from data instead of following explicit rules. Training means showing the computer many examples so it learns patterns. Classification means assigning categories to new inputs based on training. A simple ML model takes input data, compares it to trained patterns, and outputs a prediction. AI powers voice assistants, recommendation systems, and translation tools.	Students discuss where AI appears in daily life. They list examples like voice assistants, photo tagging, and spam filters.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a simple AI simulation in Scratch. Make a sprite that asks yes or no questions. Based on answers it tries to guess what the player is thinking. Use a decision tree of if-then-else blocks to narrow down possibilities. This simulates how simple AI makes decisions based on input.	Students build rule-based systems that simulate AI reasoning. They understand that simple AI follows decision trees based on user input.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Explain the difference between AI and regular programs. Describe what training means in machine learning. Give two examples of AI you use regularly. Identify one limitation of current AI systems.	Students demonstrate understanding of AI concepts. They can distinguish between programmed rules and learned patterns.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software with music extension · Note frequency chart · Song notation sheets · Tempo reference · Textbook ·

DIFFERENTIATION **App** Provide simplified AI concept cards with pictures and examples for visual learners **On** Students complete standard AI identification and basic Scratch ML extension exercises **Adv** Advanced students explore how training data quality affects AI accuracy and discuss bias... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: List three ways AI helps you on your phone. Spiral: also recall one fact from last week.

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LESSON 5.12 · DEBUG · 40 MINUTES

Capstone Game

I CAN Apply all Scratch skills to create a complete game · Plan, design, build, and test a game project · Present game with documentation and demonstration ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You have learned variables, conditions, loops, lists, clones, pen drawing, music, and AI. Now you will combine everything into one capstone project. What kind of game would you love to create that uses as many of these skills as possible?	Students brainstorm ambitious game ideas that showcase their complete skill set. They feel excited to create something they can be proud of.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine these capstone game examples from previous students. One game uses clones for enemies, variables for scoring, and loops for gameplay. Another uses pen for drawing paths and lists for level data. Identify which Scratch concepts each game uses.	Students analyze example projects and see how all concepts combine. They plan which concepts to include in their own games.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A capstone project demonstrates your complete learning. Steps include brainstorming game concept, designing mechanics on paper, creating a project plan with features list, building the game in stages starting with core mechanics, adding polish with sounds and visuals, testing with classmates and fixing bugs, documenting your code with comments, and presenting with live demonstration.	Students create project plans with timelines and feature lists. They learn to break large projects into manageable tasks.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Build your capstone game over this session and the next. Start with the core mechanic. Add scoring with variables. Include enemies or obstacles with clones. Create multiple levels with lists. Add sound effects and music. Include a start screen and game over screen. Test and balance the difficulty.	Students build complete games applying all learned concepts. They practice project management and iterative development.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your capstone game to the class. Demonstrate gameplay live. Explain three programming concepts you used and how they contribute to the game. Answer questions from classmates and teacher about your design choices.	Students present their work professionally. They articulate technical decisions and respond to questions about their creative process.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · Game design templates · Balancing worksheet · Player feedback forms · Textbook ·

DIFFERENTIATION **App** Provide game templates with core mechanics pre-built where students add features and... **On** Students create complete games from scratch following project planning guidelines **Adv** Advanced students create multi-level games with save systems, leaderboards, and adaptive... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Write a one-page game design document for your dream game. Spiral: also recall one fact from last week.

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LESSON 6.01 · DEBUG · 40 MINUTES

Online Identity

I CAN Understand the concept of online identity and digital persona · Compare online identity with real-world identity · Create a safe and responsible online profile ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	When you create a gaming account or social media profile you choose a name, a picture, and describe yourself. Is this the same person as the real you sitting in this classroom? How are your online and offline identities different?	Students discuss how they present themselves differently online and offline. They realize online identity is a constructed version of themselves.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Look at these five social media profiles. Some use real names and photos. Others use fake names and cartoon avatars. Some share personal details while others share nothing. Which profiles seem trustworthy? Which seem suspicious? Why?	Students evaluate online profiles for trustworthiness. They notice that profiles with real photos and complete information seem more genuine.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Online identity is how you present yourself on the Internet. It includes your username, profile picture, bio, and content you share. Your online identity can be the same as your real identity or completely different. Important considerations include privacy by limiting personal information, consistency by being the same person online and offline, reputation by creating content you would be proud of, and safety by protecting details like address and phone number.	Students analyze their own online identities. They evaluate what information they currently share and what they should protect.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Design your ideal online identity for an educational platform. Choose an appropriate username that does not reveal your real name. Write a bio that shares your interests without personal details. Select or create a profile image that represents you without showing your face. Explain each choice and why it is safe.	Students create safe online identities. They practice sharing personality without compromising privacy.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate these online identities for safety and appropriateness. Identify three pieces of information that should never be in an online profile. Create your own safe profile following all the guidelines we learned.	Students demonstrate understanding of online identity safety. They can identify risky information and create appropriate profiles.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch with ML extensions · AI examples · Machine learning visual demos · Ethics discussion guide · Textbook ·

DIFFERENTIATION **App** Provide profile creation templates with safety guidelines embedded in each field **On** Students create standard online profiles following privacy and safety guidelines **Adv** Advanced students research digital reputation management and create a comprehensive... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Check the privacy settings on your phone apps tonight.
Spiral: also recall one fact from last week.

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LESSON 6.02 · DEBUG · 40 MINUTES

Privacy and Data

I CAN Understand what personal data is and why it needs protection · Identify different types of data collected by apps and websites · Apply privacy settings to protect personal information ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Your favorite game app asks for permission to access your contacts, location, camera, and microphone. Why does a simple game need all this information? What could it do with your contact list and location data?	Students realize that apps collect more data than needed for their function. They wonder why games need access to contacts and location.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Open the settings on your phone. Look at the permissions for three different apps. Note what each app can access. Compare a camera app that needs camera access with a game that asks for camera access. Which permissions seem necessary and which seem suspicious?	Students discover that many apps request unnecessary permissions. They learn to evaluate whether permissions match app functionality.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Personal data includes your name, age, address, phone number, email, photos, location, contacts, and browsing history. Apps collect this data for targeted advertising, service improvement, and sometimes selling to third parties. Privacy settings let you control what data apps can access. Data protection principles include collecting only necessary data, securing data with passwords, not sharing data without consent, and deleting data when no longer needed.	Students audit their own app permissions. They identify data types that need protection and learn to adjust privacy settings.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Conduct a privacy audit of five apps on a school phone. For each app identify what data it collects, whether that data collection seems necessary, what privacy risks exist, and what settings could be changed to improve privacy. Write a report with recommendations.	Students practice real-world privacy analysis. They develop critical evaluation skills for digital privacy.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Classify each data type as sensitive or non-sensitive. Phone number, favorite color, home address, school name, age. Explain why some data needs more protection than others. Create a list of five privacy rules everyone should follow.	Students demonstrate understanding of data sensitivity levels. They create practical privacy guidelines.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Scratch software · All previous lesson materials · Project planning templates · Testing feedback forms · Presentation area ·

DIFFERENTIATION **App** Provide simplified permission review cards with pictures showing what each permission... **On** Students complete standard privacy audit exercises on provided devices **Adv** Advanced students research GDPR and Indian data protection laws and their impact on... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Tonight review the permissions for your three most-used apps. Spiral: also recall one fact from last week.

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LESSON 6.03 · DEBUG · 40 MINUTES

Cyberbullying Prevention

I CAN Define cyberbullying and recognize its different forms · Understand the impact of cyberbullying on victims · Learn strategies to prevent and respond to cyberbullying ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Imagine someone creates a fake social media account using your photo and posts embarrassing messages pretending to be you. Your classmates see it and start sharing it. How would you feel? What would you do?	Students empathize with the victim and begin discussing responses. They realize cyberbullying can happen to anyone and has serious emotional impact.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Read these five scenarios on the board. One student receives threatening messages. Another has their photo edited without permission. A third is excluded from a group chat deliberately. A fourth has rumors spread about them online. A fifth is mocked for their answers in class group. Identify which scenarios are cyberbullying and explain your reasoning.	Students analyze scenarios and identify cyberbullying behaviors. They recognize that exclusion and mocking are also forms of bullying.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Cyberbullying is using technology to hurt, embarrass, or threaten another person repeatedly. Forms include sending threatening messages, spreading rumors online, sharing embarrassing photos without consent, creating fake profiles to humiliate someone, deliberately excluding someone from online groups, and impersonating someone to damage their reputation. Impact includes anxiety, depression, loneliness, and decreased academic performance. Prevention strategies include not responding to bullies, blocking and reporting them, saving evidence, telling a trusted adult, and supporting victims.	Students learn to identify and respond to cyberbullying. They practice response strategies through role-play scenarios.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a cyberbullying prevention poster for the school. Include the definition of cyberbullying, five warning signs that someone is being cyberbullying, steps to take if you are being bullied, how to support a friend who is being bullied, and helpline numbers. Use clear language appropriate for all ages.	Students create educational materials for their school community. They apply their knowledge to help others.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Respond to these situations. You receive a mean message from an unknown number. Your friend shows you a cruel meme about another classmate. You accidentally see hurtful comments on a classmates post. Explain the appropriate response for each situation.	Students demonstrate appropriate responses to cyberbullying scenarios. They show understanding of when to report and when to intervene.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Social media profile examples · Online identity checklist · Privacy settings guide · Template profile forms · Textbook ·

DIFFERENTIATION **App** Provide simplified scenario cards with clear illustrations showing bullying and... **On** Students analyze standard cyberbullying scenarios and practice response strategies **Adv** Advanced students research cyberbullying laws and create a school policy proposal for... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Think about how you treat others in group chats and gaming. Spiral: also recall one fact from last week.

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Digital Footprint Advanced

I CAN Understand how digital footprints are created and stored · Define the long-term consequences of online actions · Take steps to manage and reduce your digital footprint ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Everything you post online stays online forever. Even if you delete a photo someone may have already screenshot it. Imagine a future employer searching your name online. What would they find? Would you be proud of what they see?	Students realize that online actions have permanent consequences. They begin to think about their digital legacy.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Search for your own name on the Internet. What appears? Now search for the name of a famous person and see how many years of content appears. Discuss how a digital footprint builds over time and how difficult it is to remove content once it is online.	Students discover their own digital footprint. They see that public figures have extensive digital histories they cannot control.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A digital footprint is the trail of data you leave when using the Internet. Active footprints are things you intentionally post like social media updates and comments. Passive footprints are data collected without your direct action like browsing history, location data, and search queries. Companies store this data on servers indefinitely. Future consequences include college admissions, job applications, and personal relationships being affected by past online behavior. Managing your footprint means thinking before posting, using privacy settings, and regularly reviewing your online presence.	Students map their own digital footprints. They categorize activities into active and passive footprints.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a digital footprint timeline of your life so far. Mark when you first went online, created your first account, made your first post, and other milestones. Estimate what data exists about you online. Write a letter to your future self about what digital footprint you want to leave.	Students reflect on their digital history and plan their digital future. They understand that they are actively creating their permanent record.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Define this person's digital footprint. What can you learn about them from their online presence? Is there anything concerning? How could they improve their digital footprint? Apply the same analysis to your own online presence.	Students develop critical awareness of digital footprint implications. They can evaluate online presence from an outsider perspective.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Phone with app settings · Privacy audit checklist · Data type classification guide · Permission review template · Textbook ·

DIFFERENTIATION

App Provide simplified digital footprint tracking sheets with icons representing common...
research data brokers and explore tools for removing personal...
On Students complete standard digital footprint analysis exercises
Adv Advanced students
SEND Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Search for your name online tonight. Spiral: also recall one fact from last week.

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LESSON 6.05 · DEBUG · 40 MINUTES

Online Reputation

I CAN Understand how online reputation affects opportunities · Build a positive digital presence through intentional content · Recover from reputation damage through corrective actions ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Two students apply for the same college scholarship. The admission officer searches both names online. Student A has a blog about science projects and coding achievements. Student B has inappropriate photos and complaints from friends. Who gets the scholarship?	students immediately recognize that online reputation affects real-world opportunities. They understand that digital presence influences how others perceive them.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine these three online profiles. Profile one belongs to a student who posts only complaints and negative comments. Profile two belongs to a student who shares school projects and volunteer work. Profile three belongs to a student who has no online presence at all. Which profile would impress a teacher or employer most?	students evaluate how different online behaviors create different reputations. They realize that having no presence is also a choice with consequences.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Online reputation is how others perceive you based on your digital presence. It affects college admissions, job opportunities, friendships, and personal relationships. Building positive reputation requires sharing achievements and interests, commenting respectfully, connecting with mentors and role models, creating original content that shows your skills, and being consistent in positive behavior. Recovery from reputation damage involves removing harmful content, creating positive content to push down negative results, apologizing sincerely if you wronged others, and building new positive history over time.	students assess their own online reputation. They create an action plan for building positive digital presence.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Audit your current online reputation. Search your name and document what appears. Identify any content that could damage your reputation. Create a plan to improve your online presence by listing five positive actions you can take this month. Start implementing one action today.	students take concrete steps to improve their digital reputation. They practice proactive reputation management.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate these online behaviors for reputation impact. Posting homework help requests. Sharing science fair project results. Leaving negative reviews of restaurants. Commenting on friends photos respectfully. Arguing with strangers online. Rate each as positive or negative for reputation.	students demonstrate understanding of how specific online actions affect reputation. They can distinguish between reputation building and damaging behaviors.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Cyberbullying scenario cards · Prevention poster materials · Helpline directory · Role-play scripts · Textbook ·

DIFFERENTIATION **App** Provide reputation audit worksheets with guided questions for analyzing search results **On** Students complete standard reputation analysis and improvement planning exercises **Adv** Advanced students create professional online portfolios showcasing their skills and... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Ask a family member what they would think if they searched your name online. Spiral: also recall one fact from last week.

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Copyright Basics

I CAN Understand what copyright protects and how it works · Identify copyrightable materials and public domain content · Apply fair use principles for educational purposes ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Your friend writes a poem and shares it online. Someone else copies it, puts their name on it, and submits it as their own work. Is this legal? Does your friend have any rights over the poem they created?	students recognize that creators have rights over their original work. They begin to understand the concept of intellectual property.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Look at these items. A published textbook. A movie. A song. A student essay. A recipe. A mathematical formula. A news photograph. Which of these are protected by copyright? Which are not? Why are some protected and others not?	students identify that creative works are protected while facts and ideas are not. They understand that copyright applies to original expression.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Copyright is a legal right that gives creators control over their original works. It automatically applies when a work is created in a fixed form. Copyright protects books, music, movies, software, photographs, and artwork. It does not protect ideas, facts, or methods. Copyright gives the creator exclusive rights to reproduce, distribute, perform, and display their work. Duration varies but generally lasts for the creator lifetime plus sixty years. Works enter public domain when copyright expires and can be freely used.	students learn copyright basics. They identify which of their own schoolwork is protected by copyright automatically.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a copyright awareness guide for our school students. Include the definition of copyright, examples of protected works, what you can and cannot do with copyrighted material, how to find copyright-free resources, and what fair use means for school projects. Use clear examples relevant to student life.	students create practical copyright guides for their peers. They apply copyright knowledge to educational context.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Is this copyright infringement. Copying a chapter from a textbook for your own use. Using a movie clip in your school presentation. Selling copies of a song you downloaded. Quoting a paragraph in your essay with credit. Explain why each is or is not infringement.	students demonstrate understanding of copyright application. They can distinguish between infringement and permitted use.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Internet search demonstration · Digital footprint timeline template · Privacy settings guide · Reflection journal prompts · Textbook ·

DIFFERENTIATION **App** Provide copyright scenario cards with clear yes and no examples for basic understanding **On** Students complete standard copyright identification and fair use analysis exercises **Adv** Advanced students explore international copyright differences and digital rights... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the textbook you use every day. Spiral: also recall one fact from last week.

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LESSON 6.07 · DEBUG · 40 MINUTES

Creative Commons

I CAN Understand Creative Commons licenses and their purposes · Identify different CC license types and their conditions · Find and use Creative Commons content legally in projects ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Copyright says you cannot use others work without permission. But what if creators want to share their work freely while still getting credit? Is there a way to give permission in advance to everyone?	students realize that copyright is restrictive and wonder if there are alternatives for sharing. They discover that Creative Commons provides flexible licensing.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Visit the Creative Commons website. Look at the six main license types. Match each symbol to its meaning. The BY symbol means attribution required. The NC symbol means non-commercial only. The ND symbol means no derivatives. The SA symbol means share alike. Find images on a CC search engine using different license filters.	students explore Creative Commons licenses visually. They practice searching for content with specific license requirements.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Creative Commons or CC provides free licenses that let creators share work with specific permissions. CC BY requires giving credit to the creator. CC BY-SA requires credit and sharing under the same license. CC BY-NC requires credit and non-commercial use only. CC BY-ND requires credit and no modifications allowed. CC0 or public domain dedication means no restrictions at all. When using CC content always follow the license requirements. Attribution should include creator name, title, source, and license type.	students practice creating proper attribution for CC content. They learn to match their project needs with appropriate license types.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Find three Creative Commons images for a school presentation on Karnataka culture. Write proper attribution for each image following CC guidelines. Explain why you chose each image and how its license affects what you can do with it. Include the attributions in your presentation credits.	students practice finding and using CC content legally. They build attribution habits for ethical content use.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Match each Creative Commons license to its requirements. Identify which licenses allow commercial use. Explain what share alike means. Create proper attribution for this CC BY-NC-SA image.	students demonstrate CC license knowledge. They can apply licensing rules correctly in their projects.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Reputation audit template · Search engine demonstration · Brand building guide · Portfolio examples · Textbook ·

DIFFERENTIATION **App** Provide CC license matching cards with simplified explanations for each license type **On** Students complete standard CC license identification and attribution exercises **Adv** Advanced students explore CC0 public domain dedication and compare CC with other open... **SEND** Touch and identify network devices with teacher guidance, use simple one-word labels

DORMITORY · PAPER

Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Search for a Creative Commons image about your favorite topic. Spiral: also recall one fact from last week.

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LESSON 6.08 · DEBUG · 40 MINUTES

Fake News Detection

I CAN Identify characteristics of fake news and misinformation · Apply verification techniques to evaluate news sources · Develop critical thinking skills for consuming online information ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	You see a news article on social media that says eating bananas cures all diseases. It has thousands of shares and comments. Does this make it true? How can you tell if news is real or fake before sharing it with others?	students realize that popularity does not equal truth. They want to learn how to verify information before believing and sharing.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine these four news items. One is from a recognized newspaper with named sources. One is from an anonymous blog with no sources. One exaggerates findings with sensational language. One presents balanced facts with multiple viewpoints. Rate each for credibility and explain your ratings.	students identify credibility indicators like named sources, balanced reporting, and professional presentation versus anonymous sources and sensational language.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Fake news is false information presented as real news. Types include propaganda with deliberate political bias, satirical content mistaken as real, misleading content with exaggerated headlines, manipulated photos or videos, and fabricated stories with no factual basis. Verification techniques include checking the source reputation, looking for named experts and sources, searching for the same story on reliable news sites, examining the date to avoid old news, checking photos using reverse image search, and reading beyond the headline to the full article.	students practice verification techniques on sample stories. They learn to pause and verify before sharing information.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Create a fake news detection checklist for our school students. Include at least ten verification steps. Test your checklist by evaluating three news stories two real and one fake. Document your verification process for each story and explain your final judgment.	students build practical verification tools. They apply systematic checking to real information evaluation.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate this news story. Check the source. Look for named experts. Search for corroborating reports. Examine the language for bias. Check the date. Write a credibility report explaining whether the story appears true or false and why.	students apply comprehensive verification methodology. They can support their credibility judgments with specific evidence.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Copyright examples · Public domain resources · Fair use guidelines · Creative Commons introduction · Textbook ·

DIFFERENTIATION	App Provide simplified verification checklists with visual cues for identifying fake news... students explore deepfake technology and analyze how AI-generated fake content...	On Students complete standard source evaluation and fact-checking exercises	Adv Advanced students explore deepfake technology and analyze how AI-generated fake content...
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Look at the news on your phone tonight. Spiral: also recall one fact from last week.	SEND Touch and identify network devices with teacher guidance, use simple one-word labels	

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LESSON 6.09 · DEBUG · 40 MINUTES

Screen Time Balance

I CAN Understand the effects of excessive screen time on health · Create a balanced schedule for screen and non-screen activities · Apply strategies for healthy technology use habits ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	How many hours did you spend looking at screens yesterday including phone, computer, and television? Research suggests teenagers should have no more than two hours of recreational screen time daily. Was your screen time above or below this recommendation?	students estimate their screen time and realize they may exceed recommended limits. They want to understand the health effects.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Keep a screen time diary for three days. Record every screen activity with start time and end time. Calculate total daily screen time. Categorize activities as educational, social, entertainment, or creative. What percentage of your screen time is educational versus entertainment?	students collect real data about their own screen habits. They discover patterns in how they spend their screen time.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Excessive screen time can cause eye strain, headaches, poor posture, sleep disruption from blue light, reduced physical activity, and decreased attention span. Balance strategies include setting daily time limits, taking breaks every thirty minutes using the twenty-twenty-twenty rule of looking at something twenty feet away for twenty seconds, creating screen-free zones like bedrooms and dining areas, replacing some screen time with physical activities, and scheduling specific times for recreational screen use.	students evaluate their own screen habits against health guidelines. They identify areas where they can reduce screen time.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Design a balanced daily schedule for a residential school day that includes study time with screens, recreational screen time limited to two hours, physical activities, social time with friends without screens, and sleep preparation without screens for one hour before bed. Explain the health reasoning behind each time allocation.	students create practical schedules that balance technology use with healthy habits. They apply health knowledge to personal planning.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Rate your current screen habits as healthy or unhealthy for each category. Eye care. Sleep habits. Physical activity. Social interaction. Academic focus. Create an action plan to improve your weakest area.	students self-assess their screen habits honestly. They create specific actionable improvement plans.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Creative Commons website · License symbol reference · CC search engine · Attribution template · Textbook ·

DIFFERENTIATION	App Provide simplified screen time tracking sheets with picture categories for different... Advanced students research digital wellness apps and present strategies for managing...	On Students complete standard screen time analysis and balance planning exercises SEND Touch and identify network devices with teacher guidance, use simple one-word labels	Adv
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Tonight set a timer for your screen use. Spiral: also recall one fact from last week.		

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Digital Wellness

I CAN Recognize signs of digital stress and technology addiction · Develop healthy relationships with technology · Practice mindfulness in digital interactions ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Have you ever felt anxious when your phone battery dies? Or felt stressed when you cannot check social media for an hour? Or stayed up late scrolling through videos unable to stop? These feelings might indicate technology is controlling you rather than you controlling it.	students recognize their own technology-related anxiety. They begin to understand unhealthy technology relationships.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Complete this digital wellness self-assessment. Rate each statement from one to five. I feel anxious without my phone. I check social media first thing in the morning. I use screens to avoid uncomfortable feelings. I lose track of time when using devices. I have arguments about screen time. Calculate your total score and identify your wellness level.	students self-assess their digital wellness honestly. They identify areas where technology use has become problematic.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Digital wellness means maintaining a healthy relationship with technology. Signs of technology addiction include feeling anxious without devices, checking phones compulsively, using screens to escape problems, neglecting face-to-face relationships, and experiencing sleep problems from late-night use. Healthy habits include setting phone-free times, practicing mindful technology use by being intentional about each interaction, taking digital detox periods regularly, engaging in offline hobbies, and maintaining real-world friendships through in-person activities.	students create personal digital wellness plans. They identify specific healthy habits to develop.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Design a digital wellness challenge for your dormitory. Create seven daily challenges one for each day of the week that encourage healthier technology use. Examples include phone-free meals, thirty-minute outdoor time, reading a physical book, and having a face-to-face conversation without phones.	students create community wellness initiatives. They practice promoting healthy habits among peers.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Identify three signs of unhealthy technology use in your own habits. Create specific strategies to address each sign. Plan a one-day digital detox and describe what you would do instead of screen time.	students recognize personal unhealthy patterns and create concrete improvement strategies. They plan practical detox experiences.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Fake news examples · Verification tools guide · Source evaluation rubric · Reverse image search demo · Textbook ·

DIFFERENTIATION	App Provide simplified wellness checklists with visual scales for self-assessment	On Students complete standard wellness analysis and planning exercises	Adv Advanced students research neuroscience of technology addiction and present findings to...	SEND Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Try a one-hour digital detox tonight. Spiral: also recall one fact from last week.			

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LESSON 6.11 · DEBUG · 40 MINUTES

Responsible Posting

I CAN Understand the impact of online posts on others · Apply the think-before-you-post framework · Create positive and constructive online contributions ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	A classmate posts a photo from a school trip. Someone comments This is so ugly and tags their friends to laugh. The original poster feels embarrassed and hurt. Was the comment necessary? Could the commenter have said nothing or been kind instead?	students recognize that online comments affect real people with real feelings. They understand that posting is a social act with consequences.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Read these five online comments on a classmate's project post. One is encouraging. One is constructive criticism. One is sarcastic and mean. One is irrelevant spam. One shares the project with others complimenting the work. Rate each comment from positive to harmful and explain your rating.	students evaluate different types of online comments. They identify which contribute positively and which cause harm.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	Every online post affects others and your reputation. Before posting use the THINK framework. Is it True? Do I need to say it? Is it Helpful? Is it Inspiring? Is it Necessary? Is it Kind? Positive posting includes congratulating achievements, offering constructive feedback, sharing useful information, encouraging struggling classmates, and celebrating others successes. Avoid posting when angry, sharing others private information, making hurtful comments, or participating in pile-on criticism.	students practice applying the THINK framework to sample posts. They learn to pause and evaluate before publishing.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Review your last ten social media posts or messages. Apply the THINK framework to each. Rate each as positive, neutral, or negative. Identify any posts you would delete or rewrite. Create a personal posting guide with rules you will follow for future posts.	students honestly evaluate their own posting habits. They create personal guidelines for responsible online communication.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Evaluate these posts using the THINK framework. A sarcastic comment about a classmates mistake. A congratulatory message on a friends achievement. Sharing someone photo without asking permission. A detailed complaint about a teacher online. Rate each and explain the appropriate response.	students apply ethical reasoning to posting decisions. They demonstrate understanding of impact and responsibility.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Screen time tracking app · Health guidelines poster · Balance schedule template · Break timer apps · Textbook ·

DIFFERENTIATION	App Provide simplified THINK framework cards with visual prompts for each question Advanced students explore online community moderation and create guidelines for managing...	On Students complete standard posting evaluation and personal guideline creation exercises	Adv Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Before you post anything on social media tonight, ask yourself each question of the THINK framework. Spiral: also recall one fact from last week.		

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LESSON 6.12 · DEBUG · 40 MINUTES

Digital Citizenship Pledge

I CAN Synthesize all digital citizenship concepts into personal commitments · Create a comprehensive digital citizenship pledge · Commit to responsible technology use and advocate for others ·

PHASE	TEACHER ACTIVITY	STUDENT ACTIVITY	ASSESSMENT CHECK
ENGAGE <i>5 min</i>	Throughout this chapter we have learned about online identity, privacy, cyberbullying, digital footprints, reputation, copyright, fake news, screen time, digital wellness, and responsible posting. How can you make sure you remember and follow all these lessons every day?	students realize that knowledge without commitment is ineffective. They want to create personal pledges to guide their digital behavior.	<i>Vocabulary pre-taught; prior ideas on board</i>
EXPLORE <i>10 min</i>	Examine these three digital citizenship pledges from students in other schools. One focuses on privacy. One focuses on kindness. One covers all major topics comprehensively. Which pledge seems most complete and practical? What topics would you add?	students evaluate existing pledges and identify gaps. They realize that comprehensive pledges cover multiple aspects of digital citizenship.	<i>Circulate; verify role rotation every 7 min</i>
EXPLAIN <i>10 min</i>	A digital citizenship pledge is a personal commitment to behave responsibly and ethically online. It should include privacy commitments about protecting personal data and respecting others privacy, safety commitments about avoiding cyberbullying and reporting harmful behavior, integrity commitments about honest posting and respecting copyright, wellness commitments about balanced screen time and healthy habits, and community commitments about supporting others and building positive digital spaces.	students draft comprehensive pledges covering all topics learned. They personalize commitments to their specific situations.	<i>Exit ticket: 2 questions at the door; check Success Criteria — if <80% correct, reteach 3 min</i>
ELABORATE <i>10 min</i>	Write your complete digital citizenship pledge. Include at least ten specific commitments covering identity, privacy, bullying, footprint, reputation, copyright, fake news, screen time, wellness, and posting. Make each commitment specific and actionable. Design an appealing presentation for your pledge. Sign it and date it. Share it with your class.	students create formal pledges demonstrating their complete understanding. They commit to specific behaviors they will follow.	<i>Peer tick against Success Criteria</i>
EVALUATE <i>5 min</i>	Present your digital citizenship pledge to the class. Explain why you included each commitment. Receive feedback from classmates on completeness and specificity. Revise your pledge based on feedback. Create an action plan for how you will remind yourself of your commitments daily.	students present their pledges confidently. They demonstrate integrated understanding of all digital citizenship concepts.	<i>Individual exit ticket (2 Q) at door</i>

RESOURCES Pledge examples · Digital citizenship review guide · Presentation materials · Accountability partner system · Textbook ·

DIFFERENTIATION	App Provide pledge templates with sentence starters for each commitment area On Students create complete pledges covering all digital citizenship topics Adv Advanced students create peer education materials teaching digital citizenship to... SEND Touch and identify network devices with teacher guidance, use simple one-word labels
DORMITORY · PAPER	Retrieval (3-2-1): Write 3 facts from today, 2 connections to previous chapter, 1 question you still have. Then do one focused task: Read your digital citizenship pledge tonight before sleeping. Spiral: also recall one fact from last week.

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